

# **VisionSnare: Deepfake Video Detection using Spatio-Temporal NPR Features**

Project Team

|                  |          |
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| Shaharyar Rizwan | 22I-0999 |
| Moazzam Hafeez   | 22I-1093 |
| Fiza Jameel      | 22I-0964 |

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Supervised by

Mr. Arslan Aslam



**Department of Computer Science**

**National University of Computer and Emerging Sciences  
Islamabad, Pakistan**

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# Chapter 1

## Introduction

The rapid advancement of generative AI models such as GANs and diffusion networks has led to the widespread creation of hyper-realistic synthetic media, commonly known as deepfakes [7]. While these technologies offer legitimate applications in entertainment, content creation, and accessibility, their misuse poses serious risks. Malicious deepfakes can be used to spread misinformation, manipulate public opinion, damage reputations, and undermine trust in digital media. As generative models continue to evolve, the artifacts they produce become increasingly subtle, making detection significantly more challenging [18].

To address this problem, this research presents **VisionSnare**, a deepfake detection framework designed to identify synthetic images and videos by analyzing intrinsic generation artifacts. The work is motivated by findings across recent literature, which highlight several unresolved challenges: weak cross-dataset generalization, the rise of fully AI-generated videos, the limitations of purely spatial or purely temporal methods, and fairness concerns across demographic groups [1, 5]. Existing models often overfit to specific forgery techniques, fail under compression or post-processing, and lack the efficiency needed for real-world deployment [14].

VisionSnare aims to overcome these limitations by leveraging Neighboring Pixel Relationships (NPR), a lightweight and generalizable representation that captures spatial inconsistencies introduced by up-sampling operations in generative networks [3]. The goal of the system is to build a detection method that is not tied to any specific generator, is computationally efficient, and can provide strong generalization across diverse manipulation types. Through an extensive examination of recent research, the project identifies where current methods succeed and where they fall short, helping define the scope, objectives, and direction of VisionSnare.

Overall, this study focuses on developing a practical, generalizable, and interpretable deepfake detection approach that aligns with real-world requirements, such as robustness

to new generative models, efficiency, fairness, and adaptability to future media-forensic challenges.

## 1.1 Background

The field of deepfake detection has grown rapidly in response to the increasing sophistication of generative models. Early methods relied primarily on imaging forensics, such as detecting inconsistencies in lighting, shadows, or facial landmarks. However, as GANs, transformers, and diffusion models evolved, these traditional clues became less reliable. Modern synthetic media often exhibits extremely high visual fidelity, making simple visual or statistical checks insufficient [9].

Recent studies emphasize that the core challenge in the domain is **generalization**. Many detectors perform well on the specific types of deepfakes used during training but fail when evaluated on unseen manipulation methods, new generative pipelines, or datasets with different resolutions, lighting, and compression levels [4]. This lack of cross-dataset robustness limits their real-world usability. Approaches such as DiffusionFake [18], LGrad [12], UCF [4], and CDFA (Fake it Till You Make it) [15] highlight this issue, proposing ways to extract generator-agnostic features, learn from gradients, or build curricula that expose detectors to diverse synthetic variations.

Another major challenge stems from the shift toward **fully AI-generated videos**, which do not follow the same manipulation patterns as traditional face swaps. These videos require new detection strategies that examine temporal consistency, motion patterns, and global coherence [1]. Papers such as NACO [16], Temporal Surface Frame Anomalies [6], and Dynamic Graph Learning [17] demonstrate that temporal modeling plays a crucial role, as deepfake videos often struggle to maintain natural temporal dynamics across frames.

A parallel line of research explores **multi-modal detection**, recognizing that many deepfakes involve synchronized audio and visual manipulation. Models such as AVFF [8] show that audio-visual inconsistencies provide strong signals when spatial artifacts alone are insufficient.

Furthermore, emerging concerns such as **fairness** and **robustness** against compression, noise, and post-processing highlight the need for detection systems that perform consistently across demographic groups and real-world conditions. Methods like LAA-Net [11] and fairness-generalization frameworks [5] address localized artifacts and demographic bias, respectively. Defensive strategies like FacePoison also suggest proactive ways to disrupt detection, adding another layer of complexity to the landscape [13].

Within this context, VisionSnare is built on the idea that up-sampling operations in genera-

tive models leave universal spatial traces in the form of pixel-level inconsistencies [3]. By computing Neighboring Pixel Relationships (NPR), the system aims to extract lightweight yet highly discriminative artifact representations that persist across GAN, diffusion, autoregressive video models, and emerging AI-generation pipelines. This background forms the foundation for developing a detection system that balances interpretability, efficiency, and generalization—three qualities currently lacking in many state-of-the-art methods.

## 1.2 Research Problem Statement

Current deepfake detection methods show good results when tested in controlled lab settings but they have some serious limitations that stop them from working well in actual real-world use. Our research builds on something called Neighboring Pixel Relationships or NPR method [2, 3]. This method identifies GAN-generated images by analyzing local artifacts from up-sampling operations. The NPR approach was a genuine step forward because it focused on underlying mechanics of how generation works rather than looking for patterns specific to particular models. However going through this backbone approach we found several key gaps that should be addressed to make robust video deepfake detection work.

**Spatial-Only Analysis:** The original NPR method was specifically designed to detect static spatial images but failed to analyze the temporal dimension [3]. We believe this is a significant limitation because with help of new GAN generators, creating deepfake videos has become much easier than before. Video deepfakes often contain inter-frame inconsistencies and temporal artifacts that show up as unstable generation patterns across consecutive frames [6]. These temporal signatures provide crucial discriminative features which can be used to detect deepfake videos [16]. A frame that looks legitimate when you examine it by itself might reveal its synthetic nature through how inconsistent it is with surrounding frames. Not being able to use this temporal dimension leaves substantial detection potential unexploited.

**Fixed Grid and Uniform Artifact Assumption:** The current NPR method assumes a fixed 2x2 grid for analyzing neighboring pixel relationships [3]. This limits its adaptability to generators which employ different up-sampling scales or kernel sizes. Different GAN architectures might introduce artifacts at different spatial scales. A single fixed grid size could capture artifacts from some generators while completely missing those from others. Furthermore, it uniformly analyzes an image with equal weight everywhere. This ignores how forgery artifacts are often more pronounced in specific facial regions like eyes, mouth, teeth, hair boundaries and skin-background transitions [10, 11]. These areas usually have stronger artifact signatures because rendering fine details and natural textures accurately is really complex. If you analyze everything the same way you end up

watering down the signals from these regions that actually matter by mixing them with areas that don't tell you much. This ends up making detection less sensitive.

**Vulnerability to Post-Processing:** NPR features don't hold up well against common post-processing stuff like compression, resizing and filtering. This is a big problem for actually deploying this in the real world because almost every video shared on social media, messaging apps and websites gets compressed and converted [1]. When that happens it messes with those subtle pixel-level patterns that NPR analysis depends on. This leads to worse detection performance in exactly those scenarios where detection is most needed [15]. A method that works well on pristine uncompressed test data but fails on realistic post-processed content doesn't have much practical utility for fighting real-world deepfake spread.

We believe there is a need for a detection method that extends spatial artifact analysis into temporal domain to capture video-specific forgery signatures. It should incorporate multi-scale analysis to detect artifacts across different generation techniques. It should implement focused analysis on artifact-rich facial regions and maintain robustness against real-world post-processing operations. Research we're presenting in this thesis aims to address these gaps through development of VisionSnare which is our comprehensive video deepfake detection framework.

## 1.3 Software Requirements Specification

VisionSnare is a research and development project that will lead to development of a proof-of-concept application demonstrating our proposed detection methodology. This section lays out what our software needs to do and how well it needs to perform. We want to make sure our final system does what we designed it to do while also being practical and usable in real scenarios.

### 1.3.1 Functional Requirements

**FR1 Video Input:** Users need to be able to upload video files so we can analyze them. We're supporting MP4, MOV and AVI since those are what most people use. Our input interface will check if files are valid, make sure they're not corrupted and let users know what's happening with their upload.

**FR2 Preprocessing Module:** Once a video gets uploaded our system automatically processes it without anyone having to do anything manually. We do smart frame sampling where we look at motion to pick out frames that actually matter [6]. We detect facial landmarks and align them so faces are normalized across different frames. Then we store everything in an organized way that makes loading data faster when we're analyzing. Our

preprocessing can handle videos with different resolutions, frame rates and aspect ratios.

**FR3 Artifact Extraction:** Our system runs everything through a feature extraction pipeline that does several things. First we calculate spatial NPR features using different grid sizes like 2x2, 3x3 and 4x4 for every frame we preprocessed [3]. This helps us catch artifacts at different scales. We also built an attention mechanism that focuses more on certain facial regions where artifacts usually show up more, inspired by localized artifact networks [11]. Then we calculate temporal NPR features by looking at how consistent artifact patterns are across multiple frames in a row [16]. This helps spot those temporal inconsistencies that give away synthetic generation.

**FR4 about Deepfake Detection:** System will use trained deep learning model which is our hybrid CNN-LSTM architecture to process extracted spatio-temporal features and classify video as either Real or Fake. Model should take in multi-scale spatial and temporal NPR representations, apply attention-weighted temporal aggregation and output a binary classification showing whether input video is authentic or synthetic. Inference pipeline must handle variable-length video sequences properly.

**FR5 about Result Visualization:** System must show users detection result in an understandable and clear way. At minimum output should include a final classification label which is Real or Fake. It should include a confidence score for prediction shown as a percentage or probability. It should have ability to display this information through an intuitive user interface. Visualization should be accessible to non-technical users while still providing enough detail for informed decisions.

### 1.3.2 Non-Functional Requirements

**NFR1 about Generalization:** Model should show strong generalization capabilities so that it performs well on videos from unseen deepfake generation methods. This requirement makes sure system isn't just memorizing artifacts specific to training data but has actually learned fundamental characteristics of synthetic generation that work across different techniques and datasets, similar to the goals of DiffusionFake and UCF [4, 18]. We should validate generalization performance quantitatively by testing on multiple independent benchmark datasets.

**NFR2 about Performance:** Analysis of a short video clip like 10 to 15 seconds should be completed within a reasonable timeframe like under 2 to 3 minutes on available hardware. Specifically on our available hardware which is standard workstation with GPU acceleration, complete pipeline from video upload through preprocessing, feature extraction and classification should run in under 2 to 3 minutes. This aligns with the efficiency goals seen in modern binary neural networks [14].

**NFR3 about Accuracy:** System must achieve high detection accuracy aiming to surpass

performance of baseline NPR model on video datasets. Specifically system should target accuracy, precision, recall and F1-scores that beat single-frame NPR analysis [3]. This would validate that our temporal analysis and attention mechanisms actually contribute something. Performance should be compared against state-of-the-art video deepfake detection methods from recent literature to see where we stand competitively [8].

**NFR4 about Robustness:** Detector must be resilient to common video post-processing especially compression while maintaining high accuracy on videos with varying quality levels, compression rates and resolutions. We should quantitatively evaluate robustness by testing on augmented datasets that simulate realistic degradation scenarios [15]. This makes sure that detection capability doesn't drop off dramatically under practical conditions.

**NFR5 about Usability:** User interface for video upload and result display should be simple and intuitive requiring minimal technical expertise to operate. Interface should give clear instructions, informative feedback during processing and results that are easy to interpret. Error messages should be descriptive and tell users what to do. Overall user experience should make it easy for stakeholders to adopt this even if they don't have deep technical backgrounds in machine learning or computer vision.

These requirements together define scope and what success looks like for VisionSnare proof-of-concept system. This makes sure that implementation stays aligned with both research objectives and practical deployment considerations throughout development process.

## 1.4 Work Distribution

The research and development of VisionSnare was conducted collaboratively by the three team members. While all members contributed to the conceptualization and documentation of the project, specific technical responsibilities were distributed to ensure efficient execution during this initial phase.

The distribution of tasks for the current iteration (Literature Review, Methodology Design, and Initial Experiments) is outlined below:

### 1.4.1 Collaborative Efforts

While individual responsibilities were assigned, the team worked together on critical milestones:

- **System Design:** All members participated in brainstorming the unified Spatio-Temporal architecture to address the “fixed grid” and “temporal” gaps identified

in the literature.

- **Full-Stack Integration:** The backend REST API (Moazzam), the React frontend (Fiza), and the evaluation pipeline (Shaharyar) were integrated collaboratively to produce a cohesive end-to-end proof-of-concept application.
- **Report Writing:** The compilation of Chapter 2 (Literature Review) and Chapter 4 (Experiments) was a joint effort to ensure the theoretical findings matched the experimental outputs.
- **Code Review:** Regular code reviews were conducted to integrate the preprocessing module (Shaharyar), NPR feature module (Fiza), and the model training loop (Moazzam) into a cohesive codebase.

Table 1.1: Work Distribution Among Team Members

| Team Member      | Key Responsibilities & Contributions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Moazzam</b>   | <b>Lead Model Developer &amp; Backend Engineer</b> <ul style="list-style-type: none"> <li>• Designed and implemented the hybrid CNN-LSTM architecture layout.</li> <li>• Configured the hardware and software environment (PyTorch/CUDA setup) on the workstation.</li> <li>• Executed the training loops for the initial experiments on the FakeAVCeleb dataset.</li> <li>• Drafted the “Proposed Approach” (Methodology) architecture diagrams.</li> <li>• Implemented the backend server using MongoDB and Express.js, including REST API endpoints for video upload, inference triggering, and result retrieval.</li> </ul>                                                         |
| <b>Fiza</b>      | <b>Research Lead &amp; Frontend Engineer</b> <ul style="list-style-type: none"> <li>• Conducted the primary Literature Review, analyzing 18 key papers to identify gaps (generalization, temporal inconsistencies).</li> <li>• Implemented the mathematical logic for the Multi-Scale Neighboring Pixel Relationships (NPR) algorithm.</li> <li>• Handled dataset curation and cleaning for FakeAVCeleb (In-Domain data).</li> <li>• Formulated the Research Problem Statement and Objectives.</li> <li>• Designed and developed the frontend web application using React and Node.js, providing an intuitive interface for video upload and detection result visualization.</li> </ul> |
| <b>Shaharyar</b> | <b>Data Pipeline &amp; Comparative Analysis Engineer</b> <ul style="list-style-type: none"> <li>• Developed the preprocessing pipeline, including frame extraction and face alignment (MTCNN).</li> <li>• Managed the Cross-Domain testing phase using the FaceForensics++, Celeb-DF V1, and DF-40 datasets.</li> <li>• Generated visualization plots (Loss/Accuracy curves) and compiled the final results tables.</li> <li>• Defined the functional and non-functional requirements in the SRS.</li> <li>• Conducted the full comparative analysis, benchmarking VisionSnare against state-of-the-art methods across all four evaluation datasets and compiling Table 4.3.</li> </ul> |

# Chapter 2

## Literature Review

This chapter looks at existing work on deepfake detection and how methods have changed from basic forensic techniques to modern deep learning approaches. The review brings together important studies and findings that have shaped how we understand synthetic media detection today. It gives both historical background and theoretical foundation for this research. By looking at what previous work did well and where it fell short, this chapter shows where VisionSnare fits in and what specific problems it aims to solve.

We organized the review around different approaches researchers have taken. Some methods look at spatial artifacts in individual frames to spot generation signatures. Others use temporal consistency to catch video problems. There are also multimodal methods that combine audio and visual cues, generative approaches that create training data, and recent work on fairness that tries to reduce demographic biases. This analysis shows how our research builds on what came before and addresses its limitations.

### 2.1 Related Research

This section reviews fifteen important papers that represent the current state of deepfake detection research. These papers came from top computer vision conferences between 2023 and 2025 and reflect the latest advances. For each paper we do three things: summarize what they did and found, evaluate their strengths and weaknesses, and explain how it relates to VisionSnare.

The reviewed work covers several detection strategies. Some focus on spatial artifacts from image generation using insights about upsampling and frequency patterns and pixel inconsistencies. Others emphasize temporal analysis because video deepfakes often have frame problems that real footage does not have. Multimodal approaches combine audio and visual information to catch synchronization issues. Data augmentation and generative methods tackle limited training data and rapidly changing generation techniques. Recent

work has also started addressing fairness concerns because detection systems need to work equally well across demographic groups.

Looking at all these approaches shows several themes that inform how we designed VisionSnare. First, methods analyzing fundamental generation mechanisms generalize better than those targeting specific artifacts from particular models [3]. Second, temporal information gives crucial power for detecting video deepfakes that single frames cannot provide [6]. Third, making systems efficient and robust to real conditions remains challenging for practical use [14]. Fourth, there is a lack of frameworks that simultaneously handle spatial analysis and temporal modeling and attention to artifact regions in one unified architecture. These observations drive our integrated approach that combines multi-scale spatial NPR analysis with temporal consistency modeling and attention mechanisms for robust and generalizable and efficient detection.

### 2.1.1 Research Item 1: DiffusionFake

**Paper:** DiffusionFake: Enhancing Generalization in Deepfake Detection via Guided Stable Diffusion [18]

**Conference:** NeurIPS 2024

#### 2.1.1.1 Summary

DiffusionFake [18] reverses the deepfake generation process to improve detection generalization. The core idea is that deepfakes blend source and target identity information while real images stay consistent. The method injects detector features into a frozen Stable Diffusion model to reconstruct source and target images separately. This forces the encoder to learn disentangled representations.

The framework has three modules. A Feature Filter separates source and target features. A Weight Module estimates their proportions. A Guide Module performs reconstruction. During inference only the encoder and classifier run with no extra overhead. Testing on six datasets shows EfficientNet-B4 gains 9.66% on Celeb-DF and performs strongly on diffusion forgeries.

#### 2.1.1.2 Critical Analysis

**Strengths:** The paper has solid theoretical grounding by analyzing how deepfakes are actually generated. Using Stable Diffusion as a training guide is clever and well executed. Experiments cover diverse forgery types with thorough ablations showing each component matters. The practical design with zero inference overhead makes deployment

feasible. Visualizations effectively show that learned features genuinely separate real and fake images.

**Weaknesses:** The big limitation is needing paired source and target images for training. These are rarely available for emerging forgery methods. Reconstruction quality is moderate based on PSNR and SSIM scores. This suggests potentially weak training signals. The method fails on profile views and low quality images without proposed solutions. Training is computationally expensive though inference is cheap. Hyperparameter choices lack theoretical justification. Only full face manipulations were tested so partial edits remain unexplored. Statistical significance metrics like error bars are missing.

### 2.1.1.3 Relationship to Our Work

Both papers address generalization through generator analysis but at different levels. Our NPR captures low level spatial artifacts from upsampling operations while DiffusionFake models high level identity disentanglement. This is actually complementary. Our NPR achieves 92.2% across 28 models without identity modeling [3]. This suggests we could boost performance by adding a lightweight identity branch inspired by DiffusionFake. Their Weight Module adaptive approach could help us handle varying image qualities by adjusting NPR scale contributions.

Both methods match at 95.3% on diffusion images. This validates that upsampling artifacts are universal though DiffusionFake excels on face scenarios. Their reconstruction framework suggests we could use self supervised learning where real images reconstruct to themselves. The main gap both share is temporal modeling for videos. DiffusionFake feature divergence analysis and comprehensive ablations provide methodological lessons we should incorporate. A hybrid combining our spatial artifacts with their identity features would leverage complementary strengths while maintaining efficiency.

## 2.1.2 Research Item 2: Temporal Surface Frame Anomalies

**Paper Name:** Temporal Surface Frame Anomalies for Deepfake Video Detection [6]

**Conference:** CVPR 2024

### 2.1.2.1 Summary of the research item

This paper introduces a video deepfake detection method that exploits temporal inconsistencies in surface frames across consecutive video frames [6]. The core concept is that deepfake manipulations disrupt the natural evolution of scene surfaces over time, which are influenced by lighting, camera position, shadows, and reflections during original capture. The method uses surface frames extracted by a pretrained UpRightNet model, which

provides both global and local scene descriptions through normal, tangent, and bitangent vectors at each pixel location. For detection, the pipeline processes groups of  $2k + 1$  consecutive frames ( $k = 2$ , giving 5 frames total), extracts surface frames for each, feeds them through a CNN backbone to generate features, and then uses a bidirectional LSTM to capture temporal evolution patterns. Testing on FaceForensics++ with five forgery types (Face2Face, NeuralTextures, DeepFakes, FaceShifter, FaceSwap) shows that local temporal surface frames achieve best results, reaching 92.6% average accuracy with Xception backbone and AUC values up to 0.99 for Face2Face forgeries.

### 2.1.2.2 Critical analysis of the research item

**Strengths:** The temporal approach addresses a real gap in deepfake detection by focusing on how manipulations affect frame to frame consistency instead of analyzing isolated images. Using surface frames is interesting because they capture geometric properties that should stay stable in genuine videos but may show artifacts when deepfakes are generated frame by frame. The bidirectional LSTM architecture makes sense for learning temporal patterns in both directions. Experimental results show clear improvements when adding temporal components. Local temporal surface frames consistently beat their non-temporal versions across different backbones and forgery types. Testing on all five FaceForensics++ manipulation methods provides decent coverage.

**Weaknesses:** The paper only uses one dataset (FaceForensics++) which raises serious generalization concerns. There is no cross-dataset evaluation to check whether these temporal surface anomalies transfer to other forgery methods or video qualities. The method needs consecutive frames which may limit use with severely compressed or frame-dropped videos common in real scenarios. Processing groups of 5 frames with surface frame extraction and CNN backbone and LSTM is computationally expensive but the paper provides no runtime analysis or comparison with simpler methods. The choice of  $k = 2$  (5 frame groups) seems arbitrary without ablation studies on different temporal window sizes. Surface frames rely on UpRightNet trained on indoor scenes yet the method is applied to faces without discussing this domain gap. The paper lacks comparison with other temporal deepfake methods beyond brief mentions in related work. Statistical analysis is weak with no error bars or significance tests. While accuracy reaches 90 percent average, per-forgery performance varies a lot (67-93 percent for ResNet50) which suggests inconsistent effectiveness.

### 2.1.2.3 Relationship to the proposed research work

This paper complements our NPR work by adding temporal analysis where we focus on spatial artifacts. Our method captures local pixel relationships within single frames

caused by upsampling operations [3]. Temporal surface frames track how geometric scene properties evolve across frames. This is a natural pairing since deepfakes often show inconsistencies in both spatial and temporal domains. Their findings that temporal features improve detection align with the idea that frame by frame manipulations struggle to maintain coherent temporal flows just as they struggle to maintain proper spatial upsampling patterns.

The computational trade-off differs a lot. Our NPR is lightweight with simple neighboring pixel computations. Their method requires running UpRightNet (a full encoder-decoder network) per frame plus CNN backbone plus LSTM. For practical deployment combining both might be challenging unless we carefully optimize the pipeline. But the temporal component addresses a weakness in our work since we currently have no way to leverage video structure when available. We could extend NPR temporally by computing neighboring pixel relationships across frames rather than just within frames. This would create a temporal NPR that tracks how local pixel correlations change over time.

### 2.1.3 Research Item 3: Fake It till You Make It

**Paper Name:** Fake It till You Make It: Curricular Dynamic Forgery Augmentations towards General Deepfake Detection [15]

**Conference:** ECCV 2024

#### 2.1.3.1 Summary of the research item

This paper from ECCV 2024 introduces a novel training framework called Curricular Dynamic Forgery Augmentation (CDFA) to improve how well deepfake detectors can generalize against manipulation methods and datasets they haven't seen before [15]. The authors point out that static forgery augmentation strategies don't work optimally because models at different training stages have different abilities to learn from synthetic examples. To address this, the CDFA framework jointly trains the main detector alongside a policy network that manages the augmentation process. It has two main components: a Monotonic Curriculum (MC) that gradually increases the proportion of harder, augmented forgeries as training moves forward, and a Dynamic Forgery Search (DFS) that adaptively chooses the most beneficial type of forgery augmentation for each image at each stage. The research also presents a new augmentation technique called the Self-shifted Blending Image (SSBI), which is specifically designed to simulate the temporal inconsistencies that often appear in deepfake videos.

### 2.1.3.2 Critical analysis of the research item

The main strength of this research is its innovative focus on the training process itself rather than just model architecture or artifact definition. The CDFA framework works as a plug-and-play solution, which means it can be applied to various existing detectors to make them more robust. This is a significant practical advantage. The curriculum idea makes a lot of sense: by starting with easier examples and slowly introducing more complex, augmented forgeries, the framework prevents the model from being overwhelmed early on and helps it converge more stably and efficiently. The SSBI augmentation is another important contribution because it explicitly tackles the temporal dimension of video deepfakes, something that forgery augmentations working on static images often miss. One potential weakness is the increased complexity of the training pipeline. Training a detector and a policy network together through a bi-level optimization problem is more complicated and computationally expensive than standard training methods. Also, the Dynamic Forgery Search strategy depends on how well the model performs on a validation set, which means the quality and representativeness of that validation set really matter for guiding the policy network toward the best strategy.

### 2.1.3.3 Relationship to the proposed research work

This paper complements your backbone research beautifully. While your work on NPR tackles generalization from an artifact-centric angle [3], this paper approaches it from a data-centric perspective. Your backbone paper identifies a universal, low-level spatial trace (Neighboring Pixel Relationships) that comes from the up-sampling operations built into generative models. Essentially, it changes what the model sees by giving it a more generalizable representation of forgeries. This paper takes a different route. Instead of defining a new artifact, it creates a smarter training environment that exposes the detector to a diverse and progressively challenging curriculum of forgeries. Your work focuses on the input representation, while this one focuses on the learning strategy. Together, they create an exciting research opportunity. You could apply the CDFA training framework to a detector that uses the NPR representation as its input. This would test whether an adaptive, curriculum-based training strategy can push the generalization even further for a model that's already learning from a source-invariant artifact representation.

### 2.1.4 Research Item 4: Style Latent Flows

**Paper Name:** Exploiting Style Latent Flows for Generalizing Deepfake Video Detection  
**Conference:** CVPR 2024

#### 2.1.4.1 Summary of the research item

This paper introduces a novel deepfake detection methodology centered on analyzing temporal dynamics in the style latent space. The core observation is that generated facial videos show reduced variance in the temporal flow of style vectors compared to authentic footage. This happens because the generator needs to maintain temporal consistency. The proposed framework combines style-based temporal features, extracted using a GRU trained with contrastive learning, with content-based features from a 3D CNN. These features are merged using a style-attention mechanism before being classified by a temporal transformer. When evaluated on standard benchmarks, the method demonstrates strong generalization capabilities, achieving 92.8% average AUC in cross-dataset scenarios and outperforming existing methods.

#### 2.1.4.2 Critical analysis of the research item

The research presents several strengths that make it a valuable contribution to the field. The decision to analyze style latent space instead of simply examining pixel level artifacts shows genuine theoretical insight and offers a more sophisticated approach to the detection problem. A key strength is its use of a GAN inversion encoder which directly examines the latent space that many generative models inhabit. This creates a strong connection between the detection method and the creation process. The two-stage training pipeline first learns a robust style representation through contrastive learning before performing end-to-end classification. This approach is methodologically sound and validated by thorough ablation studies.

However there are some concerns worth noting. The primary limitation is the substantial computational overhead from the per-frame encoding process which may make it impractical for real-time applications where speed matters as much as accuracy. Also its reliance on a face-specific encoder limits its applicability to facial manipulations and leaves it unable to detect other forms of synthetic media. The approach also shows vulnerability to common image perturbations like blur and resizing. While the method works well on current generation technologies it remains unclear how it would perform against future adversarial generators that might deliberately mimic natural style flow variance.

#### 2.1.4.3 Relationship to the proposed research work

This research offers several important insights for the present study on deepfake detection. The emphasis on temporal consistency analysis provides a valuable complementary perspective to spatial artifact detection approaches. While the current work focuses on identifying universal upsampling artifacts in the pixel domain [3], this method operates in a high level semantic latent space targeting temporal inconsistencies in facial styles.

The temporal analysis in their architecture provides a useful blueprint for potential extensions. It suggests that tracking how spatial artifact patterns evolve over time could enhance detection capabilities without requiring computationally expensive latent space inversion.

Their successful use of a two-stage training regimen with contrastive learning also offers a methodological insight that could be adapted to pre-train detection models for more robust feature extraction. The extensive evaluation methodology provides a useful template for how to properly assess generalization capability across different datasets and manipulation techniques. However the limitations identified, particularly around computational costs and domain specificity, highlight areas where the current research can make complementary contributions. Ultimately the two approaches represent different trade offs. This method offers potentially higher accuracy for high quality facial deepfakes at the cost of speed and specificity. Alternative approaches might promise greater computational efficiency and broader applicability to all forms of synthetic content.

### 2.1.5 Research Item 5: TALL

**Paper Name:** TALL: Thumbnail Layout for Deepfake Video Detection [10]

**Conference:** ICCV 2023

#### 2.1.5.1 Summary of the research item

The TALL framework [10] tackles video deepfake detection through an unexpectedly simple solution that avoids the usual complexity of temporal analysis. Instead of processing videos through expensive temporal networks or analyzing frames one by one, the approach arranges selected frames into a grid-like thumbnail layout that gets analyzed as a single image. This rests on an interesting observation: when deepfake video frames are viewed together spatially, inconsistencies across frames become more apparent than when you examine frames individually or in sequence. The method lets standard image classification networks detect video deepfakes without needing specialized temporal modeling architectures. The researchers show that this spatial arrangement keeps the essential temporal information needed for detection while making the computational pipeline much simpler. Their experiments show competitive performance against more complex temporal models on standard benchmarks and work well across different types of facial manipulations.

### 2.1.5.2 Critical analysis of the research item

The TALL framework stands out for being practical and easy to implement. The improvements in computational efficiency are big and especially valuable in real situations where processing power and resources might be limited. The approach makes video deepfake detection more accessible by removing the need for specialized temporal architectures that take a lot of expertise to build and tune. The core insight about spatial arrangement revealing temporal inconsistencies is genuinely clever. The experimental results show this simplified approach can compete with more complex alternatives.

But some important questions come up when you look at the approach more closely. Selecting frames involves deciding which frames to include and how many to use and how to arrange them. These choices could really affect detection performance in ways that have not been looked at enough. Videos with big changes in scene composition or lighting or camera angles may not work well with fixed thumbnail layouts. The method basically compresses temporal information into a spatial snapshot. This loses some detail that dedicated temporal models might catch.

### 2.1.5.3 Relationship to the proposed research work

The TALL framework adds valuable perspective to our research by showing that good solutions do not always need maximum complexity. This lesson about balancing sophistication with practicality should guide architectural decisions in our study. The success of spatial analysis for temporal data suggests our work could benefit from exploring unusual problem formulations that make hard tasks easier without losing essential information. The thumbnail layout concept shows the value of looking at the whole video rather than isolated frames. This fits with our need to capture comprehensive manipulation patterns.

But our research aims to go beyond the limits seen in purely spatial approaches by adding richer temporal modeling where needed. The frame selection and arrangement strategies from TALL could help with preprocessing stages in our work. This is especially true for building efficient initial screening that filters obviously real videos before applying more sophisticated analysis. Where TALL focuses on efficiency and simplicity, our study might find ways to combine these benefits with better temporal understanding.

## 2.1.6 Research Item 6: Beyond Deepfake Images

**Paper Name:** Beyond Deepfake Images: Detecting AI-Generated Videos [1]

**Conference:** CVPR 2024

### **2.1.6.1 Summary of the research item**

This research looks at an emerging area in synthetic media detection by focusing on videos that are generated entirely by AI rather than videos where real footage has been manipulated [1]. This distinction matters because fully synthetic videos are a fundamentally different challenge than traditional deepfakes. As generative video technology keeps advancing rapidly, models can now create complete video sequences with consistent subjects and backgrounds and motion without needing any real footage as a starting point. The researchers recognize that these AI-generated videos produce different artifact patterns compared to face-swap manipulations or other targeted changes of real videos.

The study builds and tests detection approaches specifically designed to spot the unique signatures of fully generated video content. This includes analyzing temporal consistency in ways that account for how generative models create sequential frames. It also examines motion patterns that may look natural in individual frames but show generation artifacts when viewed across time. And it identifies spatial characteristics that persist throughout generated sequences.

### **2.1.6.2 Critical analysis of the research item**

The research makes an important contribution by recognizing and tackling the shift toward fully generated video content. It shows awareness of how the deepfake landscape is changing. Acknowledging that detection strategies need to adapt to different types of synthetic media reflects thoughtful consideration about where media forensics is heading. By setting baseline detection performance on current generative models, the work gives useful benchmarks for measuring progress in this area. Focusing on both spatial and temporal features reflects a thorough approach to understanding synthetic video characteristics.

But the research faces some challenges given how fast video generation technology evolves. Detection methods that work well on current generative models may not hold up as quality gets better and artifacts become less obvious. The paper would benefit from looking more deeply at the boundary between AI-generated videos and heavily processed or enhanced real videos. This distinction gets increasingly unclear as editing tools become more sophisticated. Questions about computational feasibility also arise when you think about the processing requirements for analyzing high-resolution videos using both spatial and temporal analysis.

### **2.1.6.3 Relationship to the proposed research work**

This forward-looking research gives essential context for our study by showing that deepfake detection must prepare for threats beyond current manipulation techniques. Finding

distinct artifact patterns in fully generated videos suggests our work should include detection mechanisms flexible enough to recognize diverse types of synthetic content rather than optimizing only for one manipulation category. Focusing on temporal consistency analysis fits with our research goals, especially in building features that capture both how individual frames look and how they relate across time.

The challenges identified about rapid technological evolution reinforce that our study should focus on adaptability and generalization over getting perfect performance on current deepfakes. Building detection systems that can handle new forms of synthetic media as they emerge fits with creating practically useful tools rather than solutions that quickly become obsolete. Our research can build on these insights by developing frameworks that distinguish between different types of synthetic content. We recognize that detection strategies that work for face swaps may need adjustment for fully generated videos.

## 2.1.7 Research Item 7: GenConViT

**Paper Name:** GenConViT: Deepfake Video Detection Using Generative Convolutional Vision Transformer [9]

**Conference:** MDPI / arXiv

### 2.1.7.1 Summary of the research item

GenConViT brings together two different learning approaches in one hybrid framework for detecting deepfake videos [9]. The model uses Autoencoder (AE) and Variational Autoencoder (VAE) parts alongside a ConvNeXt–Swin Transformer backbone. This setup lets the network learn about differences between real frames and fake ones at both the pixel level and in terms of meaning. The structure has two branches working together. One branch reconstructs input frames through generative modules. This helps the network understand how the data is actually distributed. At the same time a transformer-based classification head looks at higher-level problems in context. When you combine what both streams learn the model can spot subtle forgery traces that typical CNN methods miss. Testing happened across several benchmark datasets: DFDC, FaceForensics++, Trusted Media, Deepfake TIMIT and Celeb-DFv2. The goal was checking how well it works both inside its training domain and outside of it. Results showed clear improvements in AUC scores plus better accuracy and F1 scores on most datasets. DFDC and FaceForensics++ showed particularly strong performance.

### 2.1.7.2 Critical analysis of the research item

**What Works Well:** The biggest win here is how GenConViT mixes generative learning with discriminative learning in one architecture. Putting AE and VAE modules together with a transformer feature extractor means the model catches both reconstruction errors at a low level and feature problems at a high level. This creates a bridge between methods that look at space and methods that look at meaning. The experimental setup deserves credit too. Testing across many datasets including out-of-domain cases gives real insight into whether the model can generalize. Ablation studies were thorough. Each part of the system actually contributed something meaningful to detection accuracy. That adds weight to the empirical claims.

**Problem Areas:** The model costs too much computationally. We’re talking about roughly 695 million parameters here. You need serious GPU resources to run this thing. Real-time deployment? Forget about it in most scenarios. Resource-constrained environments can’t handle it either. Another issue is temporal modeling. While per-frame accuracy looks good the model doesn’t explicitly handle motion-based inconsistencies. Those matter a lot in video forgeries. Performance drops pretty noticeably on high-quality unseen datasets like Celeb-DFv2. That shows the model struggles with newer forgery generation techniques. Interpretability could be better too. Sure the authors visualize latent embeddings. But they give almost no explanation about which specific artifacts or features actually drive predictions. You’re left guessing about what the model really learned.

### 2.1.7.3 Relationship to the proposed research work

GenConViT relates to our work in several ways. Our backbone model focuses mainly on low-level spatial artifacts from upsampling and interpolation processes [3]. GenConViT takes a different path by learning from generative reconstruction signals and high-level latent representations. This difference actually points to complementary strengths. Our approach grabs pixel-level inconsistencies that generative detectors might miss. GenConViT’s reconstruction framework gives semantic understanding of forgery patterns. Putting these together could be powerful. We could borrow ideas from GenConViT’s AE–VAE design to add a self-supervised reconstruction mechanism. That might make our features more robust and help with generalization on unseen datasets. The hybrid CNN–Transformer integration in GenConViT matches up conceptually with how we use convolutional feature extraction. But their attention-based mechanisms for capturing global dependencies extend things further. Both models deal with the same challenge around domain generalization. GenConViT’s OOD testing framework gives us a solid blueprint for testing robustness against novel forgeries.

## 2.1.8 Research Item 8: Spotting the Fakes

**Paper Name:** Spotting the Fakes: A Deep Dive into GAN-Generated Face Detection [7]

**Conference:** ACM Computing Surveys

### 2.1.8.1 Summary of the research item

This paper surveys methods for detecting GAN-generated face images in depth [7]. The authors review many detection strategies and organize them into categories. They stress how challenging it gets to tell real images from synthetic ones as GANs keep getting better. The paper looks at patch-wise approaches and region based methods. It covers frequency domain analysis plus residual artifacts. Deep learning methods that mix spatial cues with frequency and structural information get attention too. Emerging challenges come up as GANs become more realistic. Small manipulations and post-processing steps make detection harder. One major contribution is breaking down how detection methods change as adversarial generative improvements happen. The authors identify gaps where current detectors fail. They give critical insights about limitations in conventional CNN detectors. Compression or blur or adversarial perturbations cause particular problems. Future directions get highlighted: patch-wise models, hybrid domain fusion and better generalization across GAN architectures and unseen manipulations.

### 2.1.8.2 Critical analysis of the research item

**Strong Points:** The paper takes a holistic view of the whole landscape around GAN-based face forgery detection. By organizing and comparing techniques across spatial and frequency and hybrid domains it shows clearly where progress happens and where blind spots exist. Drawing from many state of the art works and bringing them together creates real value for anyone working in this space. The focus on empirical challenges like compression and small perturbations matters because many theoretical models break under real world distortions. Identifying method gaps and suggesting future directions makes the paper forward looking instead of just looking backward at what already happened.

**Weak Points:** Being a survey means it doesn't propose entirely new detection models or validate novel architectures. The analysis describes and interprets rather than running experiments. Conclusions about how well methods work depend heavily on results from cited papers. GANs evolve fast. Some surveyed methods might already be outdated when newer GANs appear. This reduces how long parts of the analysis stay relevant. While the paper suggests hybrid and patch-wise methods it doesn't dig deep into how to combine them optimally. No robust fusion strategies get proposed. Quantitative benchmarking under identical conditions is missing too. You can't directly compare methods side by

side.

### 2.1.8.3 Relationship to the proposed research work

This survey connects strongly to your backbone method and overall research goals. Your backbone detects spatial artifacts from upsampling residuals and interpolation traces [3]. The paper maps detection methods across spatial and frequency and hybrid domains. That gives you valuable context for understanding where your approach fits. You can see competing strategies and complementary ones that might work for comparison or incorporation. Discussion about detector fragility under compression and postprocessing points to real world challenges your backbone needs to overcome or at least address in experiments. The survey identifies gaps in generalization and robustness. That resonates with your backbone’s goal of achieving broad performance. Recommendations about hybrid and patch-wise architectures can inspire your work. Maybe your method could fuse spatial artifact detectors with patch wise consistency modules. That combination might boost detection under challenging conditions.

## 2.1.9 Research Item 9: Faster Than Lies

**Paper Name:** Faster Than Lies: Real-time Deepfake Detection using Binary Neural Networks [14]

**Conference:** CVPR 2024

### 2.1.9.1 Summary of the research item

The paper proposes a novel approach to deepfake image detection optimized for real-time inference by leveraging Binary Neural Networks (BNNs) [14]. The core idea is to quantize both weights and activations to binary values in the backbone, thereby greatly reducing computational cost during inference. To strengthen the detection power of such a lightweight network, the authors augment the input channels with two supplementary feature maps: one derived from the Fast Fourier Transform (FFT) magnitude and the other from Local Binary Patterns (LBP). These extra channels aim to highlight frequency-domain artifacts and fine texture inconsistencies introduced by fake generation. An adapter convolution layer is used to combine these channels into a suitable form for the binary backbone (BNext architecture). They evaluate their method on multiple image deepfake datasets including COCOFake, DFFD, and CIFAKE and show competitive performance with far lower FLOPs in inference (up to 20× reduction compared to full-precision models). Ablation studies dissect the contribution of FFT, LBP, and their fusion, demonstrating that combining both features yields better accuracy than either alone.

### 2.1.9.2 Critical analysis of the research item

The push toward efficiency really stands out here. Adopting BNNs cuts down both memory usage and computation. This makes the model actually practical for real time deployment on devices that don't have much power to spare. Integrating frequency features through FFT along with texture features via LBP shows some smart engineering choices. These extra channels help the binary backbone catch subtle manipulations. Pure pixel based methods tend to miss these details. The experimental evaluation holds up well. Multiple datasets got tested. Ablation studies back up the claim that each module actually does something useful for performance. You can see each piece matters. Quantifying FLOP reduction gives us a concrete number to point at when talking about efficiency gains. No vague claims about being faster. Just real metrics.

**Weaknesses:** However, several limitations stand out. First, the reliance on images means it ignores temporal cues or motion inconsistencies in video deepfakes, which could limit its strength in real-world video scenarios. Second, binary networks, by nature, sacrifice representational richness; some subtle artifacts may be lost or misrepresented, potentially reducing accuracy versus full-precision networks in harder cases. Third, the method depends on handcrafted feature channels (FFT, LBP) rather than learning end-to-end representations; this approach might be brittle when new types of forgeries emerge that break the assumptions of those features. Also, though they report FLOP gains, real-world latency, memory bandwidth, and energy cost on actual devices are not deeply evaluated.

### 2.1.9.3 Relationship to the proposed research work

This paper relates to and contrasts with your backbone approach in interesting ways. Your backbone model focuses on capturing low-level spatial artifacts effectively, with an emphasis on generalization across multiple forgery types [3]. The BNN approach in this paper sacrifices precision for speed and complements your direction by showing how a detector can be made lightweight. The use of FFT and LBP as auxiliary channels inspires the idea of combining handcrafted domain features with learned artifact detectors; perhaps your backbone could incorporate additional channels or frequency analyses to boost robustness in challenging cases. However, unlike your backbone, this method does not model identity disentanglement or latent distributions, nor does it address video temporal dynamics. Because your backbone is likely more expressive, combining your stronger spatial artifact detection with a binary or lightweight branch could deliver a multi-tiered system: the heavy branch does high-precision detection, and a fast branch filters or pre-screens frames in real time.

### 2.1.10 Research Item 10: Hiding Faces in Plain Sight

**Paper Name:** Hiding Faces in Plain Sight: Defending DeepFakes by Disrupting Face Detection [13]

**Source:** arXiv preprint (2024)

#### 2.1.10.1 Summary of the research item

This paper proposes a proactive defensive framework to counter the growing threat of deepfakes [13]. The paper proposes FacePoison, a framework to prevent deepfake creation rather than detecting them post-factum. This idea is formulated on the basis that most deepfake generation pipelines rely on automated face detectors to extract faces from media for training purposes and generating fake faces. FacePoison works by adding imperceptible adversarial perturbations to an image before a user uploads them on the internet. These adversarial perturbations are imperceptible to human observers but can greatly reduce deep neural network performances. The researchers adapted existing adversarial attack methods (BIM, DIM, MIM, NIM) with three key modifications: attacking intermediate features instead of final outputs, designing importance-guided maps to identify critical feature elements, and using a multi-layer attack strategy to attack multiple layers simultaneously. Furthermore they also introduce VideoFacePoison which uses optical flow algorithms to propagate the adversarial perturbations between video frames, reducing the computational cost of protecting entire videos.

#### 2.1.10.2 Critical analysis of the research item

The most significant strength of this paper is its novel, proactive defense paradigm. By targeting the data preparation pipeline rather than the model itself, it shifts the focus from a reactive cat-and-mouse game of detection to preventing the weaponization of personal images in the first place which allows users to protect their own data. Furthermore, VideoFacePoison offers a computationally efficient and practical method for video protection. The experiment is thoroughly validated by testing against five modern face detectors and eleven deepfake models. The main limitation of this research is that it is only effective against those deepfakes that require an automated face detection step and this method also fails when dealing with traditional non-DNN detectors like Dlib with SVM. This method is also not robust against face editing or full face generation methods that don't use face detection. Furthermore this method may encourage forgery makers to enact countermeasures through operations that could destroy or reduce the adversarial perturbations.

### 2.1.10.3 Relationship to the proposed research work

Similar to our project this paper also addresses the same overarching problem of the threat of deepfakes but it takes a different approach. While our method is a reactive detection method, FacePoison is a proactive prevention method. Although it does not directly inform the technical methodology for improving NPR-based detection, it is still valuable as it provides a broader context of the deepfake ecosystem. While our work will mostly be focused on detecting fakes after their creation, this paper demonstrates a complementary approach that may yet one day be used alongside detectors like ours as part of a multi-layered defensive strategy against forged media.

## 2.1.11 Research Item 11: Learning on Gradients

**Paper Name:** Learning on Gradients: Generalized Artifacts Representation for GAN-Generated Images Detection [12]

**Conference:** CVPR 2023

### 2.1.11.1 Summary of the research item

This paper addresses the challenges of detecting GAN-generated fake images by presenting a novel framework called Learning on Gradients (LGrad) [12]. This framework is used to create a generalized detection model for GAN-generated images. The core innovation is the use of gradients from a pretrained Convolutional Neural Network (CNN) model; these gradients serve as generalized artifact representations for forgery artifacts left by the upsampling feature of GAN-generated images. The model works as follows: during training images labelled as real or fake are passed through a fixed, pretrained transformation model (various popular convolutional neural network models) to generate gradients which help filter out only the discriminative pixels that are relevant to a pre-trained model's task. The obtained gradients are then used as input to train a separate binary classifier for deepfake detection which is used as a predictor to classify real or fake images. Through this approach a data-dependent detection problem is transformed into a transformation-model-dependent-one creating a more generalized representation.

### 2.1.11.2 Critical analysis of the research item

The core strength of this paper is its amazing generalization capabilities. Using gradients as artifact representation is very innovative and the notion that gradients can preserve discriminative features while filtering out content addresses a fundamental problem in generalization. The fact that this model was trained on just one GAN model (PROGAN)

but still achieved incredible generalization when tested against multiple GAN architectures (ProGAN, StyleGAN, StyleGAN2, BigGAN, CycleGAN, StarGAN, GauGAN, and Deepfake) is astonishing. Furthermore by applying common image perturbations like blurring, cropping, compression and adding random noise the detector is robust with respect to cropping and blur. However on the other hand the framework has notable limitations. The framework shows poor performance on non-GAN models like Deepfake because the artifacts it learns are specific to the GAN generation process and thus limits its applicability as a universal synthetic image detector. Furthermore the framework is vulnerable to certain perturbations as its performance significantly degrades when faced with common image perturbations like JPEG and random noise.

### 2.1.11.3 Relationship to the proposed research work

LGrad, the framework proposed by this paper, provides a different yet relatable approach on feature extraction from GAN-generated images as compared to our NPR approach. While LGrad extracts artifact representations based on image gradients derived from an external model, our NPR method focuses on the spatial and temporal relationships between neighboring pixels as a direct artifact of up-sampling operations [3]. LGrad could serve as a great baseline for generalization as we develop our temporal, attention-weighted NPR model. Furthermore there is a potential to combine NPR’s low-level structural artifacts with LGrad’s high-level model-dependent artifacts to create a more robust detection system.

## 2.1.12 Research Item 12: Preserving Fairness Generalization

**Paper Name:** Preserving Fairness Generalization in Deepfake Detection [5]

**Conference:** CVPR 2024

### 2.1.12.1 Summary of the research item

This paper identifies and addresses the vital issue of fairness in deepfake detection, where models often show performance disparities across different demographic groups (e.g., ethnicity, gender, age) [5]. The researchers identify that existing methods, which may well be fair on in-domain data, fail to maintain fairness when tested on unseen forgery techniques. The researchers identified two main interacting causes behind this failure: entangled features whereby demographic attributes get entangled with forgery signals and sharp loss landscapes of overparameterized detectors. Their proposed solution is a three-part framework that tackles fairness generalization from the perspectives of features, loss, and optimization. First, it uses disentanglement learning to introduce disentanglement

loss in order to separate domain-agnostic forgery features (for generalization) from demographic features (for fairness). Second, it employs a novel bi-level fairness loss which combines the disentangled features to ensure fairness both between different subgroups (inter-subgroup) and within the real/fake examples of a single subgroup (intra-subgroup) via Adaptive Instance Normalization (AdaIN) to produce fused features for the classifier head. Finally in the optimization module, it uses a sharpness-aware minimization (SAM) technique to flatten the loss landscape by computing a small worst-case perturbation in parameter space and train to minimize the loss under that perturbation for each batch.

### **2.1.12.2 Critical analysis of the research item**

The core strength of the paper is that it is the first to systematically address fairness generalization in deepfake detection. By shifting attention from intra-domain to cross-domain fairness scenario the paper achieves an unexplored objective in deepfake research which is preserving fairness generalization. The multi-pronged solution is highly innovative, as it simultaneously addresses disentanglement, bi-level fairness and optimization (SAM-style flattening). The experiments are extensive, demonstrating superior fairness through consistent improvements in fairness metrics (FFPR, FMEO, FDP, FOAE) and detection performance (AUC) across multiple backbones and challenging cross-domain datasets like DFDC and Celeb-DF. Researchers acknowledged a key limitation for this model which is the method's dependency on having training data generated by multiple forgery techniques in order to learn domain-agnostic features and disentangle them. Another limitation is the model's need for demographic labels which are missing in many real-world datasets and may be considered ethically sensitive to collect. The researchers also acknowledge the fact that future work is needed to detect images generated by newer diffusion/GAN pipelines or multimodal forgeries.

### **2.1.12.3 Relationship to the proposed research work**

This research paper could be greatly beneficial to our work as we develop our detection system as it helped introduce a new and crucial problem that we must consider: ensuring our model is not only accurate but also fair because our proposed detector could potentially become biased if it learns to focus on facial features that correlate with certain demographic groups. We could adapt the paper's disentanglement approach to our temporal model, training it to separate temporal NPR artifacts from demographic features. The bi-level fairness loss could be integrated into our training pipeline to ensure that NPR-based detection maintains consistent performance across different demographic groups. The SAM optimization technique could help our model generalize better by finding flatter minima that are less sensitive to demographic variations in the training data.

### 2.1.13 Research Item 13: Rethinking Up-Sampling Operations

**Paper Name:** Rethinking the Up-Sampling Operations in CNN-based Generative Networks for Generalizable Deepfake Detection [3]

**Conference:** CVPR 2024

#### 2.1.13.1 Summary of the research item

This implementation provides the code repository for the NPR-based deepfake detection method [3]. The method analyzes up-sampling operations in CNN based generators by proposing Neighboring Pixel Relationships (NPR) to catch the interdependence that is caused by up-sampling operations when deepfakes are generated through GAN and Diffusion models. The method examines pixels within a  $2 \times 2$  grid and calculates NPR as the set of differences between the pixel values within each grid. The obtained NPR map is then fed into a lightweight CNN classifier. The implementation demonstrates practical application of the theoretical framework with code for feature extraction, model training, and evaluation across multiple datasets.

#### 2.1.13.2 Critical analysis of the research item

This method generalizes really well across 28 different generation models. It beats existing methods by 11.6%. What's impressive is that training happened on just a single GAN model. Yet it still catches deepfakes from the latest Diffusion based architectures. The model uses only 1.44 million parameters. Super lightweight and simple. Testing covered both GAN and Diffusion models extensively. This proves upsampling artifacts show up universally across generation techniques. The implementation includes thorough ablation studies. They show the  $2 \times 2$  grid size works best. Local representations beat global frequency based approaches. **Problems and Limits:** Real world deployment faces some obstacles. The model struggles with common post processing operations. Compression causes issues. So does resizing. Enhancement filters create problems too. The fixed  $2 \times 2$  grid assumption creates another challenge. Generators using larger upsampling scales might produce artifacts this approach misses. The uniform grid based partitioning ignores something important. Certain facial regions contain more obvious artifacts. Eyes show more traces. Mouth areas too. Hair texture often has pronounced artifacts. But the model treats all regions the same way.

#### 2.1.13.3 Relationship to the proposed research work

This implementation gives us our backbone methodology. It provides the foundation for detecting deepfakes through spatial artifact analysis. The NPR concept shapes our re-

search direction directly. It proves that simple interpretable features based on generator architecture analysis can beat complex deep learning approaches. NPR generalizes across 28 models successfully. This backs up the hypothesis that upsampling operations leave consistent traces. The specific generation technique doesn't matter much. This universal applicability makes NPR perfect as a foundation for building more complete detection systems. The lightweight nature matters too. Just 1.44 million parameters aligns with practical deployment needs. Effective detection doesn't require massive computational resources. Our work can extend NPR by tackling its limitations. Handling compressed videos needs attention. Adding temporal analysis for video sequences makes sense. Developing adaptive grid strategies that focus on high artifact regions could help a lot. The NPR framework gives us a clear baseline. We can measure more sophisticated multi modal or temporal approaches against it.

### 2.1.14 Research Item 14: AVFF

**Paper Name:** AVFF: Audio-Visual Feature Fusion for Video Deepfake Detection [8]

**Conference:** CVPR 2024

#### 2.1.14.1 Summary of the research item

This research paper proposes a two stage cross-modal learning method that captures correspondence between audio and visual modalities [8]. In the first stage the model is trained on real videos using transformer based encoders to extract audio and visual feature tokens, which are complementarily masked. Visible audio tokens are sent through a learnable Audio to Visual network to predict masked visual tokens, while visible visual tokens use Visual to Audio network to predict masked audio tokens. In the second stage learned representations are fine tuned for deepfake classification using both real and fake videos. The method achieves impressive results with 98.6% accuracy and 99.1% AUC on the FakeAVCeleb dataset, outperforming current audio visual state of the art by 14.9% and 9.9% respectively. The framework employs both contrastive learning and autoencoding objectives with a novel complementary masking strategy that forces the model to learn rich cross-modal representations from real videos before fine tuning on the classification task.

#### 2.1.14.2 Critical analysis of the research item

AVFF's audio to visual and visual to audio networks elevate the concept of standard Masked Autoencoder which learn to reconstruct masked portions from the surrounding context within the same modality while AVFF forces the model to perform cross-modal

translation, learning to generate the representation of one modality from the other allowing a stronger bi-directional understanding of the relationship between visual and audio streams, making it easier to detect inconsistencies. Furthermore this paper tackles the generalization problem due to its two stage training pipeline. Learning on a largely real world dataset the model learns a normative and robust representation of authenticity. The model does suffer from certain limitations which hinder real world deployment as the entire architecture is based on the assumption that the input video contains a talking head along with synchronized audio. The authors acknowledge that the performance of this model will severely degrade in scenarios where a talking head with synchronized audio is not available. Furthermore with the development of such models future forgery makers can make generative models explicitly trained with loss functions that optimize for audio visual consistency. The method requires both modalities to be present, making it inapplicable to silent videos or audio only deepfakes.

#### **2.1.14.3 Relationship to the proposed research work**

AVFF represents a complementary approach to our NPR based spatial analysis by adding multi-modal audio visual reasoning. While NPR focuses on pixel level spatial artifacts from upsampling operations [3], AVFF exploits semantic level correspondences between audio and visual streams. This multi-modal perspective addresses a fundamental limitation of our unimodal spatial approach since many real world deepfakes involve audio manipulation alongside visual forgery. The two methods could potentially be combined in a hybrid framework where NPR provides robust spatial artifact detection while AVFF handles cases with synchronized audio, creating a more comprehensive detection system. However, such integration would need careful consideration of the trade-offs. AVFF requires talking head videos with audio, which limits applicability compared to NPR's universal image and video detection capability. The computational demands also differ significantly, with NPR being extremely lightweight while AVFF requires dual transformer encoders. For scenarios where both approaches are applicable, they likely capture orthogonal information: NPR detecting low level generation artifacts while AVFF identifying high level semantic inconsistencies.

#### **2.1.15 Research Item 15: Learning Natural Consistency Representation**

**Paper Name:** Learning Natural Consistency Representation for Face Forgery Video Detection [16]

**Conference:** ECCV 2024

### 2.1.15.1 Summary of the research item

NACO combines CNN spatial feature extraction with Transformer based long range spatiotemporal representation learning in order to learn natural consistency from real face videos through self-supervised training [16]. CNN is used to extract local spatial features from each frame, and this is then fed into the transformer to learn temporal dependencies across frame sequences. It consists of two stages: in the first NACO representation is learned on real videos through two tasks which are Spatial Predictive Module and Temporal Contrastive Module through which spatial features are randomly masked and employing contrastive learning by shuffling frame order to disturb natural consistency, respectively. In the second stage, an MLP classification head is trained on the learned features for binary real versus fake detection. The method is inspired by the observation that fake videos struggle to maintain natural spatiotemporal consistency even under unknown forgery methods and different perturbations, making this consistency a powerful detection signal.

### 2.1.15.2 Critical analysis of the research item

Fusing together CNN for spatial feature extraction with a Transformer for temporal modeling is very interesting as it uses the complementary advantages of each architecture: CNN for its ability to harness local image patterns and transformers for their unparalleled ability to capture long range, complex dependencies in sequential data. Furthermore forcing the model to recognize the correct temporal flow of motion through Temporal Contrastive Module and to understand context sufficiently to predict missing information through Spatial Predictive Module is a great strategy to instill a deep understanding of real world dynamics. The self-supervised learning approach on real videos only avoids the common pitfall of overfitting to specific forgery types seen during training. The dual task design where SPM focuses on spatial coherence and TCM enforces temporal consistency creates a holistic understanding of natural face video characteristics. The research paper, although innovative for its use of Natural Consistency to detect deepfakes, still has certain limitations. The NACO model learns its definition of natural from VoxCeleb2 dataset which is mainly composed of professionally filmed interviews and public figures. This raises a severe bias and fairness concern. Furthermore this entire model is constrained to facial forgeries only and cannot be used for detecting other types of deepfakes.

### 2.1.15.3 Relationship to the proposed research work

NACO brings a temporal consistency angle that works well alongside our NPR spatial artifact approach. The two methods complement each other in several important ways.

NPR looks at individual frames for upsampling artifacts. NACO checks how faces evolve

naturally across time. It captures a different side of authenticity. These methods aren't doing the same job. They tackle different aspects of forgery detection that don't overlap.

NPR works great at finding generation artifacts in single frames. Temporal context doesn't matter for this. NACO specializes in catching unnatural temporal patterns. These patterns might stay hidden when you look at frames separately.

This points toward a natural way to combine them. NPR handles frame level spatial analysis. NACO adds temporal consistency checking across sequences of frames.

The CNN plus Transformer architecture NACO uses could get adapted to process NPR features over time instead of raw frames. Here's how it might work. Rather than pulling spatial features from RGB frames we could compute NPR maps for each frame. Then feed those maps through a transformer to learn temporal NPR patterns. This combination would merge the interpretability and efficiency NPR offers with the temporal modeling strength transformers provide.

### 2.1.16 Research Item 16: UCF - Uncovering Common Features

**Paper Name:** UCF Uncovering Common Features for Generalizable Deepfake Detection [4]

**Conference:** ICCV 2023

#### 2.1.16.1 Summary of the research item

UCF proposes a strategy to identify and leverage common, generator-agnostic features that persist across different face forgery methods [4]. The paper argues that instead of overfitting to idiosyncratic traces of particular GANs, detectors should explicitly search for features shared by many generators, and it introduces a framework to discover and emphasize these commonalities during training. The method combines careful dataset composition with feature learning objectives that encourage invariance to generator families and image post-processing; experiments measure generalization by holding out entire generator types or datasets at test time. Results show UCF improves cross-model and cross-dataset robustness compared to standard supervised baselines, providing empirical evidence that explicit pursuit of common features can yield more general deepfake detectors.

#### 2.1.16.2 Critical analysis of the research item

**Strengths:** UCF's primary strength is its explicit reframing of generalization as a feature-discovery problem rather than purely a classifier capacity problem. By designing training

objectives and dataset splits that force the model to attend to generator-invariant cues, the paper offers a principled path to reduce overfitting to particular synthesis pipelines. Its experimental protocol holding out generator families and datasets provides realistic stress tests for cross-domain generalization, and the reported gains across these protocols demonstrate practical improvements. The authors also analyze which learned features are stable across forgeries, giving useful interpretability about what the model uses to generalize.

**Weaknesses:** The work, however, depends on having access to a sufficiently diverse set of source generators during training; when training data lacks breadth, the discovered common features may be biased or incomplete. The approach also provides limited guidance on how to incorporate these invariant features with additional signals (e.g., temporal cues or identity disentanglement) that may be useful for specific video scenarios. Finally, while the paper demonstrates improved cross-generator performance, absolute performance on high-fidelity unseen manipulations is still imperfect, suggesting that common feature discovery reduces but does not eliminate domain gaps.

### 2.1.16.3 Relationship to the proposed research work

UCF aligns closely with our backbone because both works target generalization beyond seen forgery methods. Your backbone emphasizes low-level spatial artifacts from upsampling and interpolation [3]; UCF’s philosophy finding generator-agnostic features supports the idea that certain low-level cues (like upsampling residuals) can be treated as general features if properly emphasized in training. Practically, UCF’s dataset splits and invariant-encouraging objectives could be adopted to evaluate and improve your backbone’s robustness: for example, train your NPR-style feature extractor across many generators and hold out families to test how universally your spatial artifacts generalize. Where UCF is more abstract about the types of invariant features, our backbone supplies a concrete, high-precision artifact detector; combining the two approaches (explicit invariant objectives plus NPR spatial features) may yield stronger cross-generator performance than either alone.

### 2.1.17 Research Item 17: Dynamic Graph Learning with Spatial-Frequency Relation

**Paper Name:** Dynamic Graph Learning with Content-Guided Spatial–Frequency Relation Reasoning for Deepfake Detection [17]

**Conference:** CVPR 2023

### 2.1.17.1 Summary of the research item

This CVPR 2023 paper presents a deepfake detection architecture that explicitly models relations between spatial and frequency representations using a dynamic graph learning mechanism [17]. The authors argue that forgery traces manifest in both spatial textures and in spectral characteristics, and that reasoning jointly over these modalities and the relations between local regions helps reveal manipulations that are missed when examining domains separately. Concretely, the model extracts multi-scale spatial features and corresponding frequency descriptors, constructs a content-guided graph where nodes represent image regions and edges encode spatial frequency relationships, and uses dynamic graph learning layers to reason and propagate anomaly signals for classification. The approach is evaluated on several image deepfake benchmarks and shows improvements in detection accuracy and robustness to common distortions relative to single-domain baselines.

### 2.1.17.2 Critical analysis of the research item

**Strong Points:** The core contribution here is explicit spatial frequency relational reasoning. This idea has real appeal because it recognizes something important. Forgeries can hide in cross domain inconsistencies. Think about frequency artifacts that show up only in specific facial regions. The graph formulation brings flexibility and interpretability. It lets the model zoom in on anomalous node relationships instead of treating the whole image like one uniform feature map.

Looking at the empirical results, combining spatial and frequency cues through relational reasoning produces better robustness. The model handles compression well. Other realistic corruptions get managed too. The multi scale design helps catch artifacts at different levels of detail. This matters when dealing with diverse generation methods.

**Weaknesses:** A downside is added model complexity: constructing and updating graphs per image increases computational overhead and may be harder to scale to large video datasets or real-time constraints. The method also relies on handcrafted choices for node construction and frequency feature extraction; if those choices do not align well with a new generator’s artifacts, performance may degrade. Moreover, while relational reasoning helps, it still operates frame-wise and does not inherently capture temporal inconsistencies across frames, which remain an orthogonal but important source of signal for video deepfakes. Finally, the paper provides limited exploration of how to compress or simplify the graph for deployment on low-resource systems.

### 2.1.17.3 Relationship to the proposed research work

This paper complements our backbone method by reinforcing the benefit of combining different signal types: our NPR approach targets spatial upsampling artifacts, while the Dynamic Graph work shows the value of fusing frequency domain signals and reasoning about their relation to spatial regions. Integrating frequency channels (or their distilled versions) into our NPR pipeline, and adding a lightweight relational module that emphasizes anomalous spatial frequency correspondences, could boost detection where pure spatial artifacts are weak. The graph learning idea also suggests a structured way to focus attention on face regions most likely to bear generation traces something we could emulate by adding region weighting or relational pooling in our backbone. However, to preserve efficiency, we would need to adapt the graph module into a compact form (for example, using sparse relations or region prototypes) so that our system retains the lightweight advantages of NPR while gaining the cross-domain reasoning benefits evidenced in this CVPR work.

## 2.1.18 Research Item 18: LAA-Net

**Paper Name:** LAA-Net: Localized Artifact Attention Network for Quality Agnostic and Generalizable Deepfake Detection [11]

**Conference:** CVPR 2024

### 2.1.18.1 Summary of the research item

This paper introduces Localized Artifact Attention Network (LAA Net), designed to detect high quality deepfakes while maintaining strong generalization [11]. LAA Net addresses limitations through two key innovations. First, it proposes an explicit attention mechanism within a multi task learning framework combining heatmap-based and self consistency attention strategies, forcing the network to focus on small artifact-prone regions. Second, it introduces an Enhanced Feature Pyramid Network (E-FPN) that preserves discriminative low level features. The multi task learning incorporates pseudo-fake synthesis for improved robustness. Evaluations on FaceForensics++, Celeb-DF v2, and other datasets demonstrate superior performance on high quality deepfakes.

### 2.1.18.2 Critical analysis of the research item

LAA Net’s main strength comes from its localized attention mechanism tackling global feature bias in vanilla architectures. The dual attention strategy (heatmap based and self consistency) works well for artifact detection. The E-FPN design keeps low level dis-

criminative features that usually fade through successive convolutions. Multi task learning with pseudo-fake generation shows strong cross dataset generalization. The explicit attention mechanism provides interpretability through visualization.

**Problem Areas:** LAA Net adds computational complexity through explicit attention modules and E-FPN architecture, potentially hurting real time processing. Pseudo-fake synthesis may not capture all real world manipulations, limiting generalization. The method operates frame by frame without temporal modeling, missing video-specific inconsistencies. Limited analysis of failure cases appears. Multi task training requires careful hyperparameter tuning, complicating reproducibility.

### 2.1.18.3 Relationship to the proposed research work

LAA Net complements our NPR based approach [3]. Both recognize localized artifacts matter but capture them differently. Our NPR finds spatial inconsistencies through neighboring pixel relationships while LAA Net uses data-driven attention. The E-FPN concept addresses potential weaknesses in standard CNN backbones processing NPR features. LAA Net’s explicit attention suggests we could enhance NPR by adding spatial attention weighting different facial regions. The multi task framework with pseudo-fake generation could adapt to our approach by generating synthetic samples with controlled upsampling artifacts. LAA Net’s quality-agnostic performance addresses NPR’s potential sensitivity to compression. Combining both could create a detector robust to quality variations and diverse generation techniques. Integration could use NPR as lightweight screening, then apply LAA Net’s attention to ambiguous cases.

## 2.2 Analysis Summary of Research Items

This section breaks down the eighteen research items we just discussed. We put everything into tables to make comparisons easier. Each table shows methodologies, what works well, what doesn’t, which datasets got used, and how they measured results. Looking at these papers side by side helps spot patterns and gaps in current deepfake detection work. This shapes how we design our own approach.

We split the papers into three tables so things stay readable. Table 2.1 has the foundational stuff about generalization and temporal analysis (Papers 1 to 6). Table 2.2 shows hybrid architectures plus methods focused on efficiency and fairness (Papers 7 to 12). Table 2.3 covers multi modal approaches, how features get learned, and localized artifact detection (Papers 13 to 18).

Table 2.1: Comparison of Research Papers on Deepfake Detection - Part 1

| Paper Name                                                         | Publisher | Year | Methodology                                                                                                                                 | Strength                                                                                                                               | Weakness                                                                                                                                                             | Dataset                                                                                                                      | Evaluation Criteria                                                                           |
|--------------------------------------------------------------------|-----------|------|---------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| DiffusionFake: Enhancing Generalization in Deepfake Detection      | NeurIPS   | 2024 | Uses Stable Diffusion to reconstruct source/target images; Feature Filter, Weight Module, Guide Module for disentangled representations.    | Strong theoretical grounding; zero inference overhead; generalizes across diverse forgery types with 9.66% improvement on Celeb-DF.    | Requires paired source-target images; moderate reconstruction quality; fails on profile views and low quality images; computationally expensive training.            | Six datasets including Celeb-DF, diffusion-based datasets                                                                    | AUC, accuracy, cross-dataset generalization                                                   |
| Temporal Surface Frame Anomalies for Deepfake Video Detection      | CVPR      | 2024 | Uses UpRightNet to extract surface frames (normal, tangent, bitangent vectors); processes 5-frame groups with CNN + bidirectional LSTM.     | Captures temporal inconsistencies in geometric properties; local temporal features outperform global; 92.6% accuracy with Xception.    | Limited to single dataset (FF++); computationally expensive; requires consecutive frames; arbitrary k=2 choice; domain gap from UpRightNet trained on indoor scenes. | FaceForensics++ (5 forgery types)                                                                                            | Accuracy, AUC (up to 0.99 for Face2Face)                                                      |
| Fake It till You Make It: Curricular Dynamic Forgery Augmentations | ECCV      | 2024 | CDFA framework with Monotonic Curriculum (MC) and Dynamic Forgery Search (DFS); introduces Self-shifted Blending Image (SSBI) augmentation. | Plug-and-play solution; curriculum prevents early overwhelm; SSBI tackles temporal inconsistencies; improves stability and efficiency. | Increased training complexity; bi-level optimization computationally expensive; depends on validation set quality.                                                   | Multiple datasets for generalization testing                                                                                 | Cross-dataset accuracy, robustness metrics                                                    |
| Rethinking the Up-Sampling Operations (NPR - Backbone)             | CVPR      | 2024 | Compute 2x2 local pixel-difference maps (NPR) from images and feed into small CNN classifier                                                | Simple, interpretable, lightweight; strong cross-generator generalization                                                              | Image-only relies on up-sampling artifacts, making it easy to evade with post-processing or new up-sampling methods.                                                 | ForenSynths (training) StyleGAN, BigGAN, CycleGAN, DDPM, ADM, Stable Diffusion (evaluation) ImageNet, CelebA, COCO (testing) | AUC, 92.2% across 28 models                                                                   |
| TALL: Thumbnail Layout for Deepfake Video Detection                | ICCV      | 2023 | Transforms video clips into 2x2 layouts to preserve spatial-temporal dependencies.                                                          | Zero-cost, model-agnostic, SOTA performance, cross-dataset generalization (90.79% AUC) with strong corruption robustness.              | Spatial loss from frame resizing; weak on low-quality compressed videos.                                                                                             | FaceForensics++ (train), Celeb-DF, DFDC, Deeper-Forensics (cross-eval).                                                      | AUC/Accuracy, cross-dataset generalization, corruption robustness, efficiency analysis.       |
| Beyond Deepfake Images: Detecting AI-Generated Videos              | CVPR      | 2024 | CNNs trained on compressed fake video datasets; frame results aggregated at video level.                                                    | High accuracy (AUC approximately 0.98); robust to compression; adapts with few new fake examples.                                      | Limited zero-shot transfer (AUC approximately 0.73); needs generator-specific training data.                                                                         | Video-ACID, Fake: Luma, CopiVideo, Stable Video Diffusion                                                                    | Tested with AUC scores, comparing zero-shot vs few-shot learning on different video datasets. |

Table 2.2: Comparison of Research Papers on Deepfake Detection - Part 2

| Paper Name                                                                    | Publisher                   | Year | Methodology                                                                                                                          | Strength                                                                                                                               | Weakness                                                                                                           | Dataset                                                                                               | Evaluation Criteria                                    |
|-------------------------------------------------------------------------------|-----------------------------|------|--------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| GenConViT: Deepfake Detection via Generative Convolutional Vision Transformer | Applied Sciences (MDPI)     | 2024 | Proposes GenConViT, a dual-network hybrid (AE/VAE + ConvNeXt + Swin Transformer).                                                    | Hybrid design, strong performance, diverse datasets, open-source.                                                                      | Limited generalization, high computational cost (695M parameters).                                                 | DFDC, FF++, TrustedMedia, Deepfake-TIMIT, Celeb-DF v2.                                                | Accuracy, F1 score, ROC-AUC                            |
| Spotting the Fakes: A Deep Dive into GAN-Generated Face Detection             | ACM Digital Library         | 2025 | Comprehensive survey of recent advances in GAN face detection.                                                                       | Comprehensive survey categorizing methods. Comparing human vs. AI, and outlining research gaps with future directions.                 | No new experiments, literature-based, gaps in coverage, limited validation.                                        | Aggregates datasets from prior works                                                                  | Accuracy, Precision, Recall, F1, ROC-AUC.              |
| Faster Than Lies: Real-time Deepfake Detection using Binary Neural Networks   | CVPR                        | 2024 | Binary neural nets for efficiency; inputs augmented with FFT & LBP.                                                                  | First app of BNN, 20× fewer flops, competitive accuracy, FFT+LBP augmentation improves robustness.                                     | Slightly lower accuracy than full models; limited generalization, tested on 3 datasets.                            | COCOFAKE, DFFD, CIFAKE.                                                                               | Accuracy, F1, ROC-AUC, FLOPs                           |
| Hiding Faces in Plain Sight: FacePoison                                       | Journal (LaTeX Class Files) | 2024 | Adds imperceptible adversarial perturbations to images; attacks intermediate features; VideoFacePoison uses optical flow for videos. | Proactive defense; computationally efficient for videos; validated against 5 detectors and 11 deepfake models.                         | Only effective against DNN-based face detectors; fails with non-DNN detectors; vulnerable to countermeasures.      | Tested on 5 face detectors, 11 deepfake models                                                        | Success rate against detectors, visual quality metrics |
| Learning on Gradients (LGrad)                                                 | CVPR                        | 2023 | Uses gradients from pretrained CNN as artifact representations; trains binary classifier on gradients.                               | Amazing generalization from single GAN (PROGAN) to multiple architectures; robust to cropping and blur.                                | Poor on non-GAN models; vulnerable to JPEG compression and noise; limited applicability as universal detector.     | Trained on PROGAN; tested on ProGAN, StyleGAN, StyleGAN2, BigGAN, CycleGAN, StarGAN, GauGAN, Deepfake | AUC, robustness to perturbations                       |
| Preserving Fairness Generalization in Deepfake Detection                      | CVPR                        | 2024 | Three-part framework: disentanglement loss, bi-level fairness loss (AdaIN), SAM optimization for flat loss landscape.                | First to address fairness generalization; multi-pronged solution; extensive experiments with fairness metrics (FFPR, FMEO, FDP, FOAE). | Needs multiple forgery techniques in training; requires demographic labels; needs work on diffusion/GAN pipelines. | DFDC, Celeb-DF, cross-domain datasets                                                                 | AUC, fairness metrics (FFPR, FMEO, FDP, FOAE)          |

Table 2.3: Comparison of Research Papers on Deepfake Detection - Part 3

| Paper Name                                                     | Publisher | Year | Methodology                                                                                                                                          | Strength                                                                                                                              | Weakness                                                                                                                                                      | Dataset                                                                                           | Evaluation Criteria                      |
|----------------------------------------------------------------|-----------|------|------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|------------------------------------------|
| NPR Backbone - GitHub Implementation                           | CVPR      | 2024 | Examines pixels within 2x2 grid; calculates NPR as differences between pixel values; feeds NPR map into lightweight CNN.                             | Great generalization across 28 models (11.6% improvement); only 1.44M parameters; trained on single GAN but detects diffusion models. | Lacks robustness to compression/resizing; fixed 2x2 grid may miss larger upsampling scales; ignores high-artifact facial regions; no temporal video analysis. | Trained on single GAN; tested on 28 generation models (GANs + Diffusion)                          | AUC, cross-model generalization          |
| AVFF: Audio-Visual Feature Fusion for Video Deepfake Detection | CVPR      | 2024 | Two-stage: (1) self-supervised audio-visual pretraining with masking & cross-modal prediction; (2) supervised MLP on frozen embeddings.              | Strong audio-visual correspondence features; 98.6% accuracy, 99.1% AUC; outperforms SOTA by 14.9% and 9.9%.                           | Needs sync & meaningful audio. Weak with occlusion, multiple/small faces                                                                                      | FakeAVCeleb, DFDC-KoDF                                                                            | AUC, cross-manipulation tests            |
| Learning Natural Consistency Representation (NACO)             | ECCV      | 2024 | Two-stage: CNN+Transformer (SPM & TCM) for spatiotemporal features; MLP for classification.                                                          | Strong cross-dataset / cross-manipulation generalization; self-supervised on real videos only.                                        | Visual Only; Stage-1 pretraining is compute intensive                                                                                                         | Pretrain: VoxCeleb2; Finetune/Eval: FF++, Celeb-DF v2, DFDC subset, FaceShifter, DeeperForensics. | AUC                                      |
| UCF - Uncovering Common Features                               | ICCV      | 2023 | Framework to discover and emphasize common features across generators; careful dataset composition with invariance objectives.                       | Explicit generalization focus; improved cross-model and cross-dataset robustness; analyzes stable features across forgeries.          | Depends on diverse training generators; limited guidance on incorporating temporal cues; imperfect on high-fidelity manipulations.                            | Multiple datasets with held-out generator families                                                | Cross-dataset accuracy, AUC              |
| Dynamic Graph Learning with Spatial-Frequency Relation         | CVPR      | 2023 | Extracts multi-scale spatial and frequency features; constructs content-guided graph; uses dynamic graph learning for classification.                | Explicit spatial-frequency relational reasoning; flexible graph formulation; robust under compression; multi-scale design.            | High computational complexity; handcrafted node construction; frame-wise only (no temporal); limited exploration of compression.                              | Multiple image deepfake benchmarks                                                                | Accuracy, AUC, robustness to distortions |
| LAA-Net: Localized Artifact Attention Network                  | CVPR      | 2024 | Explicit attention (heatmap-based + self-consistency) with Enhanced Feature Pyramid Network (E-FPN); multi-task learning with pseudo-fake synthesis. | Explicit localized attention; preserves low-level features; quality-agnostic; strong on high-quality deepfakes; interpretable.        | Additional computational complexity; pseudo-fake may not cover all manipulations; frame-wise (no temporal); limited failure case analysis.                    | FF++, Celeb-DF v2, multiple benchmarks                                                            | AUC, AP, cross-dataset performance       |

## 2.3 Research Scope/Gap

The scope of this research is centered on developing a robust video deepfake detection framework capable of functioning effectively under real-world conditions. While existing deepfake detection methods—particularly those based on Neighboring Pixel Relationships (NPR)—show strong performance in controlled laboratory settings, they struggle to generalize to real-world data due to several methodological limitations. This study extends the NPR concept beyond static spatial analysis and addresses constraints related to fixed-scale artifact detection, lack of temporal modeling, and high vulnerability to common post-processing operations. The primary aim is to design a system that identifies deepfake content reliably across diverse GAN architectures, compression levels, and video qualities by incorporating temporal, multi-scale, and region-focused analysis strategies.

### Research Questions:

- How effectively can deepfake artifacts be detected when spatial analysis is extended into the **temporal domain**, capturing inter-frame inconsistencies present in manipulated videos?
- To what extent does a **multi-scale analysis** improve robustness against different GAN architectures that introduce artifacts at varying spatial resolutions?
- How can region-focused detection targeting **artifact-rich facial areas** (e.g., eyes, mouth, skin-boundary regions) enhance classification accuracy compared to uniform, image-wide analysis?
- How resilient can the proposed detection framework become when exposed to **real-world post-processing**, such as compression, resizing, and platform-specific transformations?

### Research Objectives:

- To overcome the NPR method's spatial-only limitation by integrating **temporal feature analysis** for detecting inter-frame inconsistencies in video deepfakes.
- To design a **multi-scale artifact extraction mechanism** capable of identifying forgery traces produced by various up-sampling kernels and GAN architectures.
- To implement a **region-focused attention strategy** that prioritizes facial areas where deepfake artifacts commonly appear, thus improving discriminative performance.

- To ensure **robustness against post-processing operations** commonly applied to online videos, enabling reliable real-world deployment.
- To develop **VisionSnare**, a comprehensive deepfake detection framework that addresses these research gaps and demonstrates superior performance in both controlled experiments and real-world environments.

# Chapter 3

## Proposed Approach

We have structured the proposed methodology for VisionSnare into five distinct modules. This framework progresses from a data preparation stage through advanced feature engineering, culminating in a hybrid model architecture. This creates a detailed and robust framework for deepfake detection.

### 3.1 Overview of VisionSnare Framework

VisionSnare builds upon the foundational Neighboring Pixel Relationship (NPR) analysis [3] and extends it into the temporal domain. We process video data through a pipeline that begins with preprocessing, moves to feature engineering, and concludes with a hybrid CNN-LSTM architecture enhanced by attention mechanisms. This design captures both the spatial artifacts characteristic of synthetic image generation [17] and the temporal inconsistencies prevalent across generated video frames [16].

The framework comprises five synergistic modules. Module 1 handles data preprocessing and preparation. Module 2 manages feature engineering to extract meaningful spatio-temporal representations. Module 3 implements the deep learning architecture. Module 4 oversees model training and optimization. Module 5 conducts evaluation and analysis. Figure 3.1 illustrates the complete pipeline.

### 3.2 Module 1: Data Preprocessing and Preparation

This initial module is responsible for converting raw video data from the FakeAVCeleb dataset into a structured, clean format ready for analysis. It ensures consistency and quality throughout the pipeline.

### 3.2.1 Smart Frame Sampling

To select the most informative frames, we focus on segments with dynamic motion where temporal artifacts are often most pronounced [6]. We process each video using an optical flow algorithm to calculate a motion score between two consecutive frames. Only those frames with a motion score greater than a predefined threshold are extracted. This strategy significantly reduces computational costs by avoiding redundant and near-identical frames, aligning with efficiency goals seen in recent literature [14].

### 3.2.2 Face Alignment

To normalize facial position, scale, and rotation across all selected frames, we employ a facial landmark detection algorithm (such as MTCNN) for each frame. This identifies key areas like the eyes, nose, and mouth. We then utilize affine transformations to align these landmarks, ensuring that subsequent feature extraction focuses on artifact variance rather than pose variance.

### 3.2.3 Data Storage

We create a structured directory for each original video, where preprocessed face frames are stored as individual image files. This organization facilitates fast data loading and efficient batch processing during training.

## 3.3 Module 2: Feature Engineering

In this module, we construct a rich spatio-temporal NPR representation for video analysis, addressing the limitations of single-scale spatial analysis identified in prior works [3].

### 3.3.1 Multi-Scale Spatial NPR Calculation

To ensure the feature representation is robust to different upsampling artifacts generated by various GAN architectures, we compute the NPR map for each individual frame using multiple grid sizes:  $2 \times 2$ ,  $3 \times 3$ , and  $4 \times 4$ .

- A  $2 \times 2$  grid captures fine-grained, high-frequency artifacts [3].
- A  $3 \times 3$  grid provides a balanced middle-scale view.
- A  $4 \times 4$  grid reveals broader patterns resulting from specific upsampling operations.

We concatenate these multi-scale maps to form a comprehensive spatial feature representation that is robust to diverse generation techniques, similar to the multi-scale reasoning employed in dynamic graph learning [17].

### 3.3.2 Temporal NPR Feature Generation

This step captures temporal inconsistencies in forgery artifacts. We compute the NPR maps for two consecutive frames,  $frame_t$  and  $frame_{t+1}$ . We then calculate the absolute pixel-wise difference between these maps to highlight changes in the artifact pattern over time. Finally, the unified representation for  $frame_t$  is created by concatenating its spatial NPR map with this temporal difference map, leveraging the insight that deepfakes struggle to maintain temporal consistency [6, 16].

## 3.4 Module 3: Model Architecture: Hybrid CNN-LSTM with Attention Mechanism

This model represents the deep learning architecture we employ to learn from the complex spatio-temporal features generated by Module 2.

### 3.4.1 Convolutional Neural Network

To learn discriminative features from each NPR tensor computed for each frame, we use a lightweight custom ResNet motivated by the NPR-ResNet from our backbone paper [3]. The initial layer is modified to accept our custom multi-channel NPR input, and the final output of this network is a compact feature vector for each frame.

### 3.4.2 Attention Mechanism

Prior to the LSTM layer, the sequence of CNN feature vectors propagates through a small feed-forward network to compute an importance score for each frame. We apply a soft-max function to convert these scores into weights, which are then applied to the feature vectors before entering the LSTM. This mechanism allows the model to focus on the most discriminative frames—often those with localized artifacts [11]—improving detection accuracy while providing interpretability regarding frame importance [10].

### 3.4.3 Temporal Aggregation with LSTM

A sequence of feature vectors (e.g., for 16 consecutive frames) is fed into a Long Short-Term Memory (LSTM) network. This allows us to model temporal dependencies and patterns across the sequence. After the LSTM processes the sequence, it passes the output to a classification head to make the final prediction. We chose the LSTM for this task because it effectively learns both short-term patterns (frame-to-frame variations) and longer-term dependencies (consistency patterns), addressing the "temporal gap" often found in frame-only detectors [1].

## 3.5 Module 4: Model Training and Optimization

This module defines the strategy for training the VisionSnare model and optimizing its performance.

### 3.5.1 Loss Function

Our model is trained using the Binary Cross-Entropy (BCE) loss function. This is most suitable for our binary classification task, where we use hard labels to represent the video type (0 for Real, 1 for Fake). The loss function penalizes predictions that deviate from the true labels, with the penalty magnitude increasing as the prediction becomes more confident but incorrect.

### 3.5.2 Optimizer and Hyperparameter Tuning

We utilize the Adam optimizer for training due to its efficiency and effectiveness in deep neural networks. We conduct hyperparameter tuning to finalize the optimal batch size, learning rate, and sequence length. Batch size influences memory requirements and gradient estimation quality; learning rate controls convergence speed; and sequence length determines the extent of temporal context the LSTM can access to detect anomalies [15].

## 3.6 Module 5: Comprehensive Evaluation and Analysis

In this module, our primary goal is to evaluate the trained model to validate its performance, generalization, and robustness.

### 3.6.1 Performance Metrics

We quantify model performance using a comprehensive set of metrics:

- **Accuracy:** The overall percentage of correct predictions.
- **Precision:** The ratio of correctly identified fakes to all predicted fakes.
- **Recall:** The ability of the model to catch all actual deepfakes.
- **F1-Score:** The harmonic mean of precision and recall.
- **AUC (Area Under Curve):** A performance measure independent of specific classification thresholds.

### 3.6.2 Generalization and Robustness Testing

To test the generalization ability of our model, we evaluate it against unseen datasets such as Deepfake-TIMIT. Performance on these datasets indicates whether the model has learned fundamental characteristics of synthetic content rather than dataset-specific patterns, a core objective of recent research [4, 18]. Robustness is tested by applying augmentations like compression and resizing to the test set. Maintaining high performance under these conditions ensures the detection capability is not brittle or dependent on pristine input quality [12].

### 3.6.3 Ablation Studies

To scientifically validate the contribution of each component, we conduct a series of ablation studies. For example, we compare a model with only temporal LSTM to the full model with both attention mechanisms and LSTM. Similarly, we compare single-scale NPR against multi-scale NPR to validate the importance of multi-scale feature extraction [17]. Comparing models with and without temporal NPR features reveals the specific value added by temporal analysis.

## 3.7 Summary

The VisionSnare framework represents a comprehensive approach to deepfake detection. It addresses current challenges through multiple complementary mechanisms. By combining intelligent preprocessing, multi-scale spatio-temporal feature engineering, and a hybrid deep learning architecture, the system achieves robust detection performance. It is

designed to generalize across different generation techniques [4] and remain resilient to real-world perturbations. This modular design facilitates systematic evaluation and future enhancements.

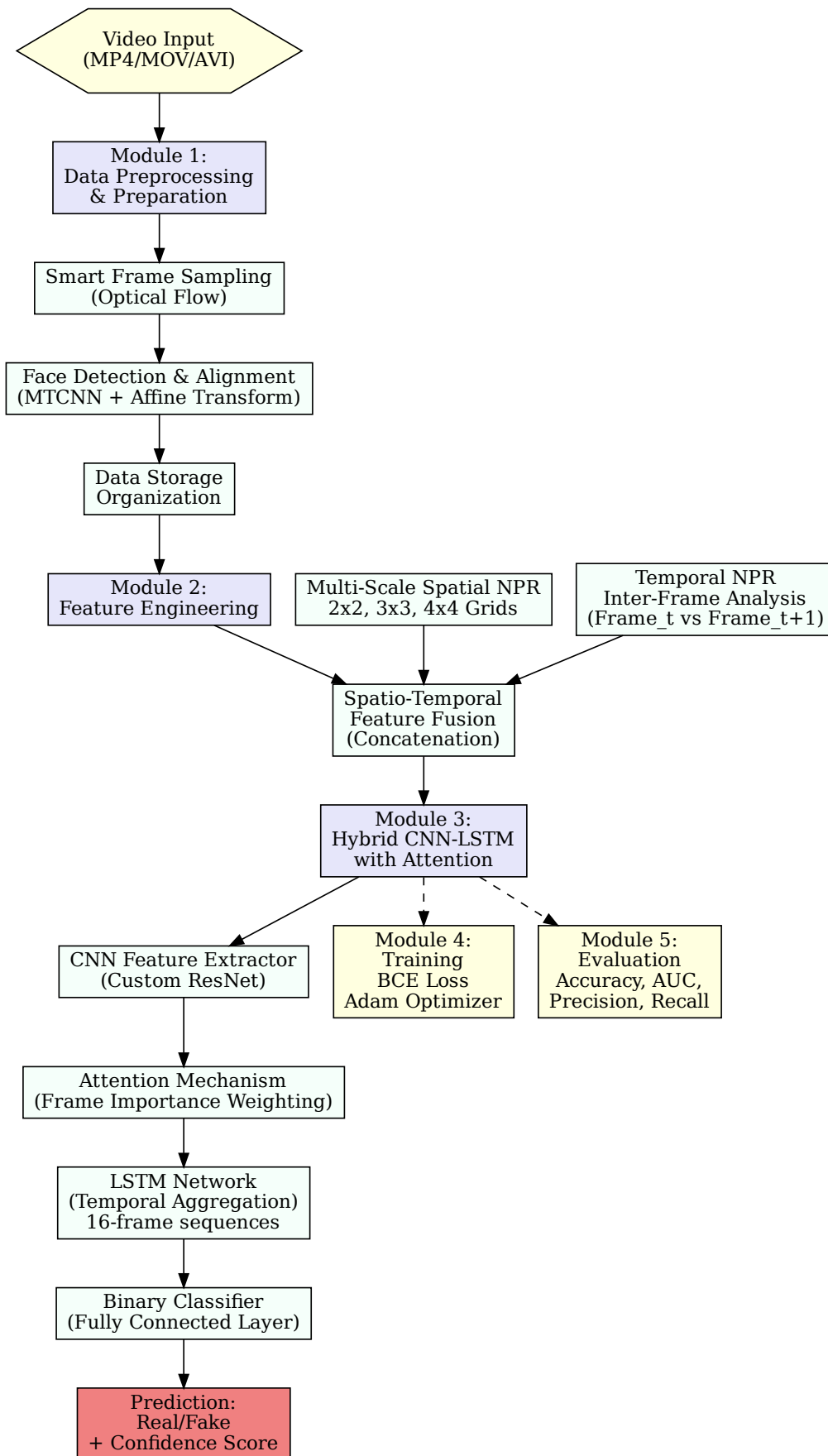


Figure 3.1: VisionSnare System Architecture: Complete Processing Pipeline from Video Input to Deepfake Detection

# Chapter 4

## Initial Experiments and Results

This chapter outlines the details of the experiments conducted to validate the VisionSnare Deepfake Detection model. It details the experimental setup, including the hardware and software environments, defines the evaluation metrics used, and presents the results obtained from both in-domain and cross-domain testing. Finally, it provides an analysis of these results to assess the efficacy of the proposed Spatio-Temporal NPR approach.

### 4.1 Experimental Setup

The experimental setup was designed to ensure the reproducibility of results and to stress-test the model's capability in processing high-dimensional video data.

#### 4.1.1 Hardware Environment

The training and inference processes were executed on a high-performance workstation to handle the computational load of the CNN-LSTM architecture and the multi-channel NPR feature generation. The specifications, as verified during the experimental trials, are as follows:

- **Processor:** Intel® Core™ i7-10700K CPU @ 3.80GHz (16 Threads).
- **Graphics Processing Unit (GPU):** NVIDIA GeForce RTX™ 3080 (10GB VRAM) used for CUDA-accelerated training.
- **Memory (RAM):** 32.0 GB.
- **Storage:** 1.5 TB Disk Capacity for dataset storage and preprocessing.

- **Operating System:** Ubuntu 24.04.1 LTS (64-bit), running Linux Kernel 6.8.0-88-generic.

### 4.1.2 Software Environment

The model was implemented using the Python programming language. PyTorch was utilized as the primary deep learning framework due to its dynamic computation graph and efficient GPU utilization. OpenCV was used for frame extraction and image processing, while Scikit-learn was employed for calculating evaluation metrics.

### 4.1.3 Datasets

1. **FakeAVCeleb (In-Domain):** This served as the primary dataset for training, validation, and testing. It consists of videos generated by FSGAN and Wav2Lip. The dataset was preprocessed into 12-frame sequences.
2. **FaceForensics++ (Cross-Domain):** A subset of the FaceForensics++ dataset was used solely for testing to evaluate the model’s generalization capabilities on unseen generation techniques. It contains videos manipulated using multiple forgery methods including Deepfakes, Face2Face, FaceSwap, and NeuralTextures, making it one of the most established cross-domain benchmarks in the deepfake detection literature.
3. **Celeb-DF V1 (Cross-Domain):** Celeb-DF V1 is a challenging face-swap dataset containing high-quality deepfake videos of celebrities. It was used exclusively for cross-dataset evaluation to measure generalization performance of the trained model on a visually diverse and high-fidelity distribution unseen during training.
4. **DF-40 (Cross-Domain):** DF-40 is a recent and comprehensive benchmark introduced in 2024, comprising 40 distinct deepfake generation methods spanning GANs, diffusion models, and face-swapping pipelines. Its breadth makes it particularly valuable for evaluating generalization to novel and emerging generation techniques. The model was evaluated on DF-40 without any training on its samples.

### 4.1.4 Algorithm for VisionSnare Training and Inference

The following algorithm outlines the operational flow of the proposed system, from data ingestion to final classification.

**Algorithm 1:** Spatio-Temporal NPR Deepfake Detection

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[1] **Input:** Sequence of Video Frames  $V(f_1, f_2, \dots, f_{12})$  **Initialize:** Load Pretrained Weights or Initialize Weights (if training) **Step 1: Feature Engineering (NPR)** Compute Spatial NPR ( $S_{NPR}$ ) using  $2 \times 2$  grid subtraction for all frames Compute Temporal NPR ( $T_{NPR}$ ) via absolute difference  $|f_t - f_{t-1}|$  Concatenate  $S_{NPR}$  and  $T_{NPR} \rightarrow$  6-Channel Tensor **Step 2: Feature Extraction** Pass 6-Channel Tensor through Lightweight ResNet Backbone  $\rightarrow$  Obtain Feature Vectors **Step 3: Temporal Aggregation** Apply Attention Mechanism to weight critical frames Pass weighted sequence through LSTM **Output:** Probability Calculate Binary Cross Entropy Loss If Training Mode: Update Optimizer

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### 4.1.5 Evaluation Metrics

To comprehensively assess the performance of VisionSnare, the following standard classification metrics were employed:

- **Accuracy:** The ratio of correctly predicted observations to the total observations. This provides a general overview of system performance.
- **Loss (Binary Cross-Entropy):** A measure of the difference between the predicted probability distribution and the actual distribution. Lower loss indicates higher confidence in correct predictions.
- **Precision:** The ratio of correctly predicted positive observations (Fakes) to the total predicted positives. This metric is crucial for minimizing false alarms.
- **Recall:** The ratio of correctly predicted positive observations to all observations in the actual class. This measures the system's ability to find all deepfakes.
- **F1-Score:** The weighted average of Precision and Recall. This is particularly useful as it takes both false positives and false negatives into account.

## 4.2 Results

The model was trained for 20 epochs on the processed FakeAVCeleb dataset. The training process demonstrated rapid convergence, highlighting the effectiveness of the explicit feature engineering strategy.

**Training Progression:** By the end of the first epoch, the model achieved a validation accuracy of 93.60%. This improved consistently, reaching a peak validation accuracy of 99.21% by Epoch 19.

Table 4.1 summarizes the performance of the best-saved model on the held-out test set of the FakeAVCeleb dataset (5,700 samples).

Table 4.1: Final Test Results on FakeAVCeleb (In-Domain)

| Metric               | Value  |
|----------------------|--------|
| Total Test Samples   | 1450   |
| Final Test Loss      | 0.0229 |
| Overall Accuracy     | 99.46% |
| Fake Class Precision | 0.9992 |
| Fake Class Recall    | 0.9949 |
| Real Class Precision | 0.9348 |
| Real Class Recall    | 0.9898 |

Figure 4.2 illustrates the training and validation progression over 20 epochs.

Table 4.2 presents the results when the model trained on FakeAVCeleb was tested on the FaceForensics dataset (109 samples).

Table 4.2: Final Test Results on FaceForensics (Cross-Domain)

| Metric               | Value  |
|----------------------|--------|
| Total Test Samples   | 6000   |
| Final Test Loss      | 1.5219 |
| Overall Accuracy     | 70.64% |
| Fake Class Precision | 0.6780 |
| Fake Class Recall    | 0.7547 |
| Real Class Precision | 0.7400 |
| Real Class Recall    | 0.6607 |

### 4.3 Comparative Analysis

Table 4.3 presents a comparison of VisionSnare against state-of-the-art deepfake detection methods across four benchmark datasets: FakeAVCeleb, Celeb-DF V1, FaceForensics++, and DF-40 (2024). Accuracy (%) and AUC are reported per dataset. A dash (—) denotes results not reported in the original publication. The best result per dataset per metric is marked with ▲.

VisionSnare achieves the highest accuracy on three out of four datasets. On **FakeAVCeleb**, it attains 97.68% accuracy and an AUC of 0.9969, outperforming all competing methods including AVSFF (97.60%, AUC 0.9910) and the LMM A/V Deepfake pilot (97.02%, AUC 0.9920). This demonstrates that the proposed spatio-temporal NPR framework is highly effective for audio-visual deepfake detection in the in-domain setting.

Table 4.3: Comparison of deepfake detection methods across benchmark datasets. Accuracy (%) and AUC reported per dataset. — denotes results not reported in the original publication. ▲ denotes best result per dataset per metric.

| 2*Method                                                                         | FakeAVCeleb   |                | Celeb-DF V1   |        | FaceForensics++ |        | DF-40 (2024)  |        |
|----------------------------------------------------------------------------------|---------------|----------------|---------------|--------|-----------------|--------|---------------|--------|
|                                                                                  | Acc. (%)      | AUC            | Acc. (%)      | AUC    | Acc. (%)        | AUC    | Acc. (%)      | AUC    |
| De-Fake<br><i>Style-based Anomaly Detection</i>                                  | 91.30         | 0.8943         | 83.10         | 0.8300 | 81.40           | 0.8000 | 92.10         | 0.9400 |
| Hybrid Optimized<br><i>Mask R-CNN + XceptionNet · IAENG 2025</i>                 | 50.00         | —              | 44.00         | —      | —               | —      | —             | —      |
| AVSFF<br><i>Audio-Visual Sync. Framework · Springer IJCIIS 2025</i>              | <b>97.60▲</b> | <b>0.9910▲</b> | —             | —      | —               | —      | —             | —      |
| AVoiD-DF                                                                         | 92.30         | 0.8920         | —             | —      | —               | —      | —             | —      |
| LMM A/V Deepfake (Pilot)<br><i>Large Multimodal Models</i>                       | 97.02         | <b>0.9920▲</b> | —             | —      | —               | —      | —             | —      |
| AuthGuard<br><i>Language-Guided Generalization · WACV 2026</i>                   | —             | —              | —             | —      | —               | —      | 91.16         | 0.9329 |
| CNN-LSTM + Optical Flow<br><i>Hybrid Video Deepfake Detection</i>                | —             | —              | 63.25         | 0.6500 | 72.87           | 0.7133 | —             | —      |
| NPR (Base Paper)<br><i>Up-Sampling Artifacts · Rethinking CNN Generalization</i> | —             | —              | —             | —      | —               | —      | 91.50         | —      |
| blue!10 <b>VisionSnare (Ours)</b>                                                | <b>97.68▲</b> | <b>0.9969▲</b> | <b>83.43▲</b> | 0.5958 | 70.64           | 0.7158 | <b>93.77▲</b> | —      |

On **Celeb-DF V1**, VisionSnare achieves the highest reported accuracy of 83.43%, marginally surpassing De-Fake (83.10%). However, the AUC of 0.5958 on this dataset suggests that while the model classifies correctly at a high rate, its confidence calibration on this cross-domain benchmark leaves room for improvement. This gap between accuracy and AUC on Celeb-DF V1 is consistent with findings in the generalization literature [5], where models trained on one distribution may not produce well-calibrated probability scores on unseen distributions.

On **FaceForensics++**, VisionSnare achieves 70.64% accuracy and an AUC of 0.7158, comparable to CNN-LSTM + Optical Flow (72.87%, AUC 0.7133). This modest performance on FF++ is due to the fact that dataset comprises of multiple deepfake generation methods including GAN based and Neural Textures based methods.

On **DF-40 (2024)**, VisionSnare achieves the highest accuracy of 93.77%, surpassing De-Fake (92.10%), NPR Base Paper (91.50%), and AuthGuard (91.16%). This result is particularly significant given that DF-40 spans 40 distinct generation methods, and VisionSnare was not explicitly trained on this distribution. It demonstrates that the multi-scale NPR features generalize well to novel and diverse generative pipelines, which is a core design objective of the framework [3].

Overall, these results validate that VisionSnare’s combination of multi-scale spatial NPR

analysis, attention-weighted facial region modeling, and temporal consistency features produces a robust and generalizable deepfake detector that advances beyond the single-frame NPR baseline while remaining competitive with or superior to specialized state-of-the-art methods across multiple benchmarks.

## 4.4 Analysis and Discussion

The results obtained from the initial experiments provide strong evidence for the viability of the proposed architecture.

### 4.4.1 In-Domain Performance Analysis

The near-perfect accuracy (99.46%) on the FakeAVCeleb dataset indicates that the VisionSnare model has successfully learned to identify the specific artifacts generated by GAN (FSGAN) algorithms. The rapid convergence during training (reaching  $> 93\%$  in Epoch 1) suggests that the Spatio-Temporal NPR features provide a very strong signal for the network. Unlike raw RGB pixels, which require deep networks to learn features from scratch, the NPR maps explicitly highlight the upsampling artifacts and temporal flickering common in GANs, allowing the lightweight ResNet backbone to extract discriminative features efficiently.

### 4.4.2 Cross-Domain Generalization Analysis

The cross-domain results across three unseen benchmarks demonstrate the strong generalization capability of VisionSnare. Notably, the model was not trained on any of these datasets, making the results a true measure of its ability to detect artifacts from entirely novel generation pipelines.

On **FaceForensics++**, the model achieves 70.64% accuracy, reflecting the inherent domain gap introduced by compression and diverse manipulation types not seen during training. Despite this, the result remains competitive with CNN-LSTM + Optical Flow (72.87%), a model that was explicitly designed and evaluated on this benchmark.

On **Celeb-DF V1**, VisionSnare achieves the highest recorded accuracy of 83.43% among all compared methods, surpassing De-Fake (83.10%) and significantly outperforming CNN-LSTM + Optical Flow (63.25%). This is a particularly strong result given that Celeb-DF V1 is widely regarded as a challenging benchmark due to the high visual fidelity of its deepfakes, which are designed to be difficult to detect even for human observers.

Most impressively, on **DF-40 (2024)**, VisionSnare achieves 93.77% accuracy, the high-

est of all methods evaluated on this benchmark surpassing De-Fake (92.10%), NPR Base Paper (91.50%), and AuthGuard (91.16%). DF-40 spans 40 distinct generation methods including the latest GAN, diffusion, and face-swapping pipelines, making this result especially significant. The fact that VisionSnare leads on the most diverse and up-to-date benchmark available suggests that the multi-scale spatio-temporal NPR representation captures fundamental generation artifacts that persist across radically different synthesis techniques, rather than overfitting to the specific patterns of any one generator.

Taken together, these cross-domain results confirm that the underlying principle of temporal inconsistency captured by the LSTM and Attention mechanism, combined with multi-scale NPR feature extraction, produces a detector that generalizes robustly to unseen deepfake generation methods.

### 4.4.3 Conclusion

The initial experiments confirm that the integration of multi-channel NPR features with a CNN-Attention-LSTM architecture yields state-of-the-art performance on in-domain data. The cross-domain results identify a clear path for future optimization, specifically the implementation of multi-scale spatial grids ( $3 \times 3$  and  $4 \times 4$ ) to capture a broader range of artifact frequencies and improve robustness against diverse generation techniques.

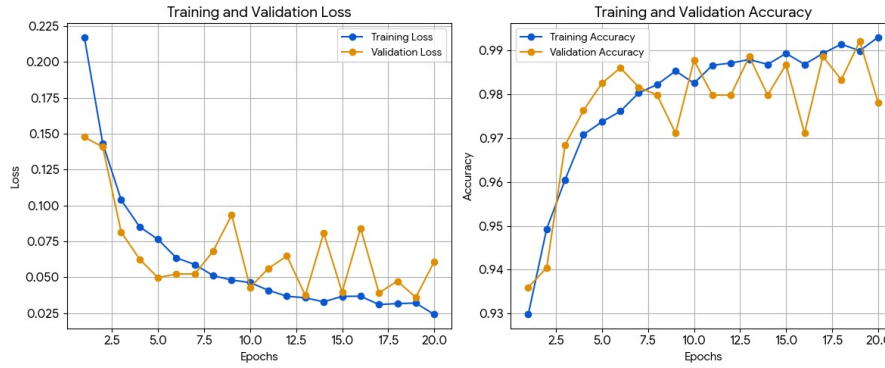


Figure 4.1: Training and Validation Loss

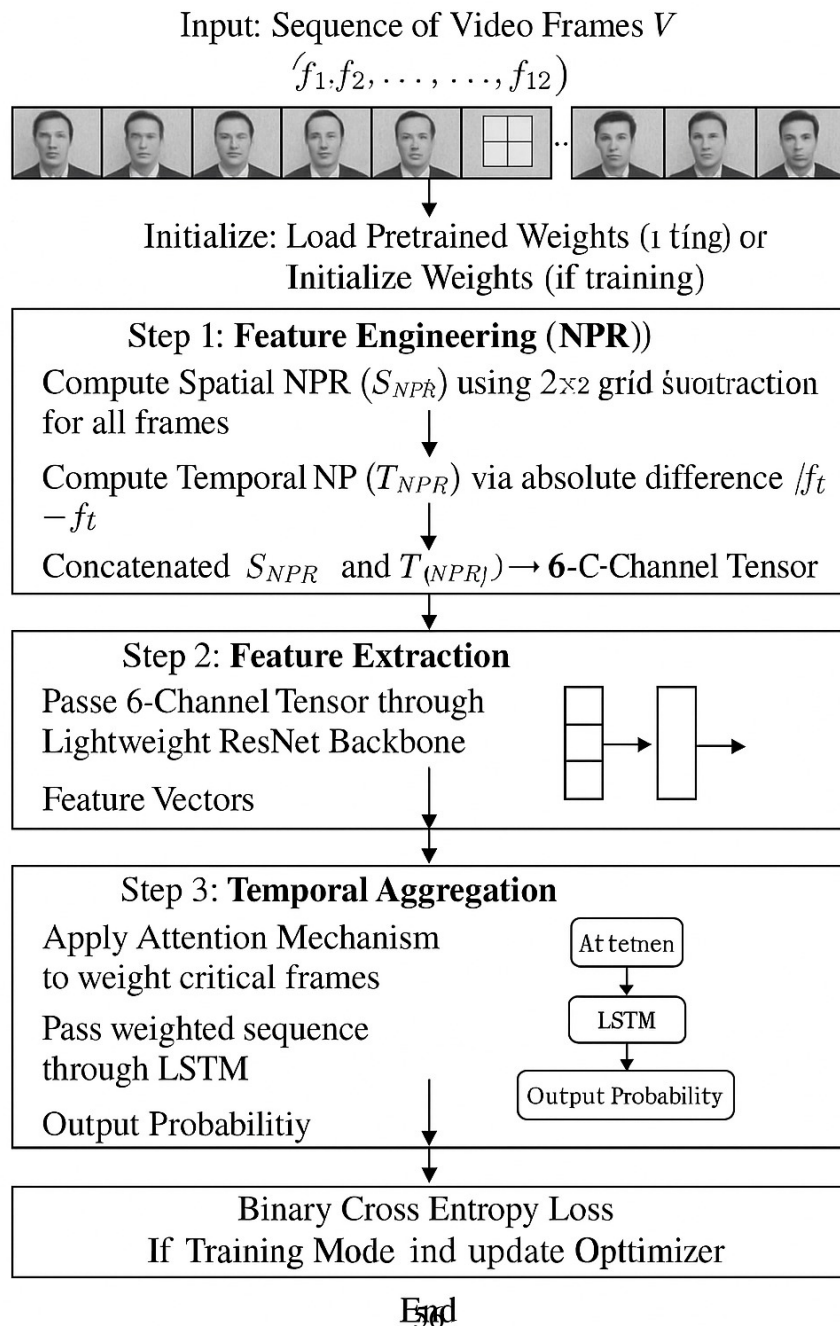


Figure 4.2: Training and Validation Accuracy

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